

Probabilistic Combinatorics

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Contents

1. Extremal Combinatorics	3
1.1. Property B	9
1.2. Schütte Tournaments	18
1.3. The Erdős–Ko–Rado Theorem	23
1.4. Bollobás Set-Pair Systems	27
1.5. Application: Saturated Graphs	28
2. Linearity of Expectation	32
2.1. Expectation and the First Moment Principle	32
2.2. Sum-Free Sets	34

1. Extremal Combinatorics

Lecture 1

Question 1.0.1. How large can a structure be if it satisfies certain properties?

15 Sep.

Example 1.0.2 (Mantel, 1907). How many edges can a graph in n vertices have if it doesn't contain a triangle?

To show the answer is $f(n)$, we need to do two things.

- (1) Show there is a triangle-free graph on n vertices with $f(n)$ edges.
- (2) Prove that any n -vertex graph with $> f(n)$ edges must have a triangle.

Theorem 1.0.3 (Mantel, 1907)

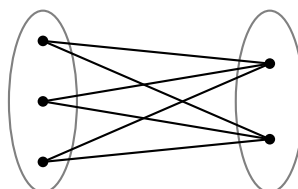
The largest n -vertex triangle-free graph has $\left\lfloor \frac{n^2}{4} \right\rfloor$ edges.

Remark 1.0.4. The complete graph K_n on n vertices has $\binom{n}{2} \approx \frac{1}{2}n^2$ edges.

proof of Mantel's theorem.

- (1) Bipartite graphs are triangle-free. Thus, we can split the vertices into two parts, take all edges between the parts. Suppose first part is of a vertices and the other one is of $n - a$ vertices.

$A, |A| = a \qquad B, |B| = n - a$



$K_{a, n-a}$: all edges between the two parts

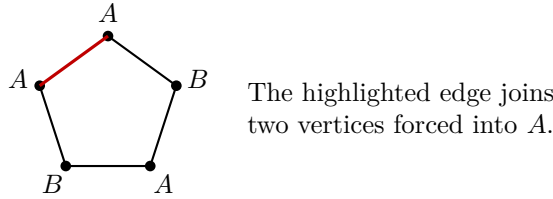
Note that this construction is triangle-free because given any three vertices, two fall in same part, and thus there is no edge between them. Note that the number of edges is $a(n - a) \leq \left\lfloor \frac{n^2}{4} \right\rfloor$, where the equality holds iff $a = \left\lfloor \frac{n}{2} \right\rfloor$ or $\left\lceil \frac{n}{2} \right\rceil$.

Note 1.0.5. Every bipartite graph is triangle-free, but a triangle-free graph need not be bipartite. For example, C_5 consists of the five edges of a pentagon, with no diagonals, so it contains no triangle.

To see why C_5 is not bipartite, suppose that its vertices could be partitioned into two sets A and B , with every edge joining different sets. Put the top vertex in A ; we can always exchange the names of the two sets to do this. Moving clockwise around the pentagon, each vertex must belong to the opposite set from the previous vertex. Thus the five vertices are forced to belong to

$$A, \quad B, \quad A, \quad B, \quad A.$$

But the last vertex is adjacent to the first vertex, and both are in A , a contradiction. Hence no bipartition exists.

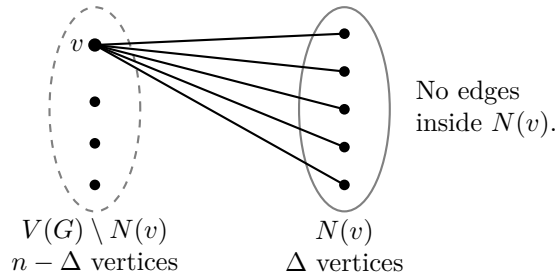


The labels A, B indicate the proposed parts, not vertex names. The red edge is an ordinary edge of C_5 , highlighted only to show the contradiction; this is not an edge-coloring problem.

(2)

Claim 1.0.1. If G is an n -vertex, triangle-free graph, then $e(G) \leq \left\lfloor \frac{n^2}{4} \right\rfloor$.

proof of claim. Let $v \in V(G)$ be a vertex of maximum degree Δ . Since G is triangle-free, the neighborhood of v is independent (no edges).



Thus, every edge contains at least one vertex of $V(G) \setminus N(v)$, where this set includes v itself. Thus,

$$e(G) \leq \sum_{u: u \sim v} \# \{ \text{edges containing } u \} = \sum_{u: u \sim v} \deg(u) \leq \sum_{u: u \sim v} \Delta = \Delta(n - \Delta) \leq \left\lfloor \frac{n^2}{4} \right\rfloor.$$

■

Remark 1.0.6. Up to isomorphism, the unique n -vertex triangle-free graph with $\lfloor n^2/4 \rfloor$ edges is the balanced complete bipartite graph

$$K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}.$$

To prove uniqueness, suppose that G attains this bound. As in the proof above, choose a vertex v of maximum degree Δ , and write

$$A = N(v), \quad B = V(G) \setminus N(v).$$

Then $|A| = \Delta$, $|B| = n - \Delta$, and A is independent because G is triangle-free. The inequalities in the proof become

$$\left\lfloor \frac{n^2}{4} \right\rfloor = e(G) \leq \sum_{u \in B} \deg(u) \leq \Delta(n - \Delta) \leq \left\lfloor \frac{n^2}{4} \right\rfloor.$$

Since the first and last quantities agree, every inequality must be an equality. Each equality gives a structural restriction:

- (i) **There are no edges inside B .** Let $e(B)$ denote the number of edges with both endpoints in B . Since there are no edges inside A , the degree sum over B counts every edge between A and B once and every edge inside B twice. Thus

$$\sum_{u \in B} \deg(u) = e(G) + e(B).$$

Equality $e(G) = \sum_{u \in B} \deg(u)$ forces $e(B) = 0$. Both A and B are therefore independent, so G is bipartite with these parts.

- (ii) **Every possible edge between the parts is present.** Each $u \in B$ has $\deg(u) \leq \Delta$. Equality

$$\sum_{u \in B} \deg(u) = |B|\Delta$$

forces $\deg(u) = \Delta$ for every $u \in B$; otherwise the sum would be strictly smaller. All neighbors of u lie in A , which has exactly Δ vertices. Hence every vertex of B is adjacent to every vertex of A , giving $G = K_{\Delta, n-\Delta}$.

- (iii) **The two parts are as equal in size as possible.** Finally, $\Delta(n - \Delta) = \lfloor n^2/4 \rfloor$. Using

$$\Delta(n - \Delta) = \frac{n^2 - (2\Delta - n)^2}{4},$$

we obtain $2\Delta - n = 0$ when n is even, and $2\Delta - n = \pm 1$ when n is odd. Thus the part sizes are $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$.

Conversely, this balanced complete bipartite graph is triangle-free and has exactly $\lfloor n^2/4 \rfloor$ edges, as shown in the construction. This proves both extremality and uniqueness up to relabeling vertices. ■

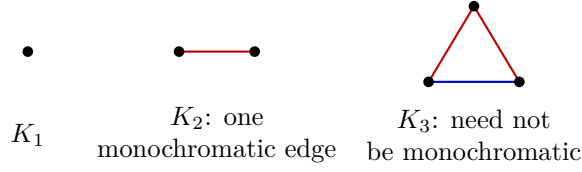
Definition 1.0.7. The **Ramsey number** $R(k)$ is the smallest n s.t. every red-/blue-coloring of the edges of K_n contains a monochromatic copy of K_k .

Theorem 1.0.8 (Ramsey, 1930)

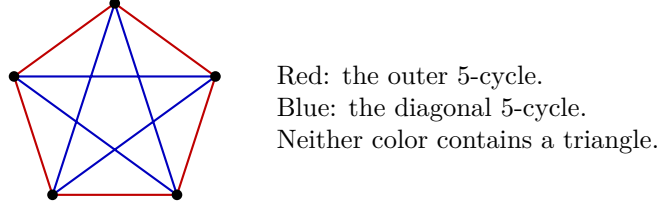
$R(k)$ is finite.

We can simply show $R(1) = 1$, $R(2) = 2$, and for $R(3)$, it is a little harder to show $R(3) = 6$.

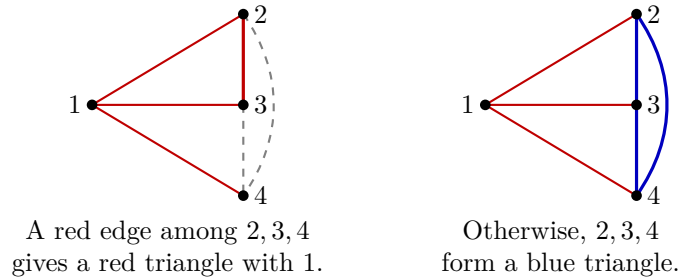
Claim 1.0.2. $R(1) = 1$, $R(2) = 2$, and $R(3) = 6$.



We can first show $R(3) > 5$ by giving an example:



We can also show $R(3) \leq 6$ by pigeonhole principle: Consider K_6 with vertices indexed by 1, 2, 3, 4, 5, 6, we know there exists 3 edges incident to vertex 1 are of same colors, WLOG, say vertices 2, 3, 4 are of red. Then we can show that no matter how we color $\{2, 3\}$, $\{3, 4\}$, $\{2, 4\}$, there must be a monochromatic triangle.



Only the relevant edges are shown; vertices 5, 6 and all other edges can be ignored. Hence every red-/blue-coloring of K_6 has a monochromatic triangle, so $R(3) = 6$.

Theorem 1.0.9 (Erdos-Szekeres, 1935)

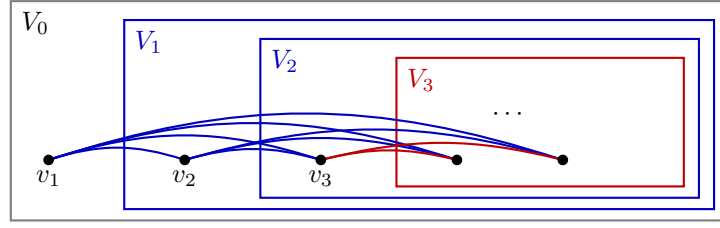
$$R(k) \leq 4^k.$$

Proof. Let $k \geq 2$, and consider any red-/blue-coloring of $K_{2^{2k}}$. Set $V_0 = V(K_{2^{2k}})$. We successively choose vertices $v_1, v_2, \dots$, colors $c_1, c_2, \dots$, and nested sets $V_0 \supseteq V_1 \supseteq V_2 \supseteq \dots$.

At step i , choose $v_i \in V_{i-1}$. Among the edges from v_i to $V_{i-1} \setminus \{v_i\}$, at least half have the same color, say c_i . Let V_i be the set of endpoints of these edges. Then every edge from v_i to V_i has color c_i , and

$$|V_i| \geq \left\lceil \frac{|V_{i-1}| - 1}{2} \right\rceil \geq 2^{2k-i}.$$

The last inequality follows inductively from $|V_0| = 2^{2k}$; the ceiling accounts for removing v_i before keeping a color class.



Keep a monochromatic neighborhood at each step.

In particular, for $j < \ell \leq i$, we have $v_\ell \in V_j$, so the edge $v_j v_\ell$ has color c_j . Also, every edge from v_j to V_i has color c_j .

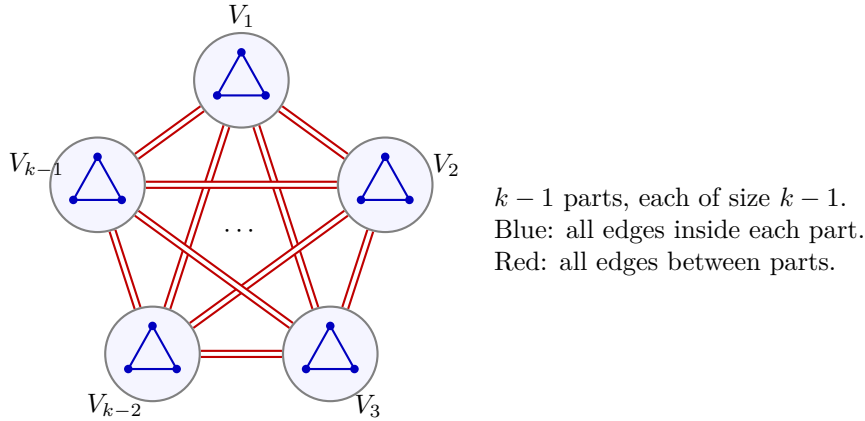
Repeat until $i = 2k$. This is possible since $|V_{2k-1}| \geq 2$, and hence V_{2k} still contains at least one vertex. We obtain

vertex	v_1	v_2	v_3	$\cdots$	v_{2k-1}	v_{2k}
color	c_1	c_2	c_3	$\cdots$	c_{2k-1}	c_{2k}

By the pigeonhole principle, at least k of the colors $c_1, \dots, c_{2k}$ are equal. Choose indices $i_1 < \dots < i_k$ with this common color. For any $r < s$, the edge $v_{i_r} v_{i_s}$ has color c_{i_r} , so these k vertices form a monochromatic K_k . Thus $R(k) \leq 2^{2k} = 4^k$. The case $k = 1$ is immediate. ■

Remark 1.0.10. More careful counting gives $R(k) \leq \binom{2k-2}{k-1}$.

For the lower bound, we can use a smart construction to show $R(k) > (k-1)^2$: For $k \geq 2$, partition the vertices of $K_{(k-1)^2}$ into $k-1$ parts $V_1, \dots, V_{k-1}$, each of size $k-1$. Color all edges inside a part blue and all edges between different parts red.



The diagram is schematic: each circle represents an entire blue K_{k-1} , and each red connection represents all edges between two parts. A blue clique must lie in a single part, so it has at most $k-1$ vertices. A red clique contains at most one vertex from each part, so it also has at most $k-1$ vertices. Thus this coloring has no monochromatic K_k , proving $R(k) > (k-1)^2$.

Theorem 1.0.11 (Erdos, 1947)

$$R(k) > (\sqrt{2} - o(1))^k = 2^{(\frac{1}{2} - o(1))k}.$$

Proof. Take the complete graph K_n on n vertices. Coloring each edge red or blue with probability $\frac{1}{2}$, independently.

The bad events, say E_S , is the event that set S of k vertices whose edges are all the same color. Then

$$\mathbb{P}(E_S) = 2 \cdot \left(\frac{1}{2}\right)^{\binom{k}{2}}.$$

Then

$$\begin{aligned}\mathbb{P}(\text{coloring is bad}) &= \mathbb{P}\left(\bigcup_{S \in \binom{V}{k}} E_S\right) \leq \sum_{S \in \binom{V}{k}} \mathbb{P}(E_S) = \binom{n}{k} 2 \left(\frac{1}{2}\right)^{\binom{k}{2}} < 2n^k 2^{-\frac{k(k-1)}{2}} \\ &= 2 \left(\frac{n}{2^{\frac{k-1}{2}}}\right)^k.\end{aligned}$$

If $n < (1-c)2^{\frac{k-1}{2}}$ for some constant $c > 0$, then

$$\mathbb{P}(\text{bad coloring}) \leq 2(1-c)^k \rightarrow 0,$$

and thus

$$\mathbb{P}(\text{coloring has no monochromatic } K_k) > 0,$$

i.e. $\exists$ a coloring of K_n without monochromatic K_k . Hence, we must have $R(k) > n$. ■

Theorem 1.0.12

$$2^{\frac{1}{2}k} \lesssim R(k) \leq 2^{2k}.$$

Definition 1.0.13 (Extremal problem). An **extremal problem** is a problem to find the size of a largest structure with a desired property.

To solve such problem, we should

- (1) show a structure of large size with given property. (lower bound)
- (2) prove that larger structures can't have property. (upper bound)

Definition 1.0.14 (Probabilistic method). To use **probabilistic method** to solve a problem, you should construct a probability space on the set of structures, then show that the desired property has positive parts, which shows the existence of a structure with such property.

Lecture 2

Recall . Erdős's lower bound on $R(k)$ colored the edges of K_n red and blue uniformly at random and applied the union bound over all k -sets of vertices: if

$$\binom{n}{k} 2^{1-\binom{k}{2}} < 1,$$

then with positive probability no copy of K_k is monochromatic, so $R(k) > n$.

That argument never used that the colored objects are the edges of a complete graph. All it needed was a family of sets, the edge sets of the copies of K_k , none of which may end up monochromatic. This lecture studies that coloring problem in its own right.

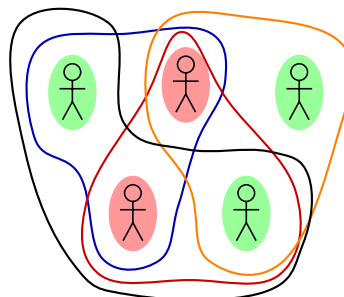
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§1.1. Property B

§1.1.i. Two-Colorability of Hypergraphs

Example 1.1.1 (Chess Olympiad). A national team travels to the Chess Olympiad. Its members fill the following roles.

- Team manager
- Team captains
- Women
- Endgame specialists
- Opening experts
- e4 players
- d4 players
- Sport psychologist



A member can hold several roles, e.g. a team captain who is also an endgame specialist, and a role is usually held by several members. The e4 and d4 players are those whose usual first move is 1.e4 and 1.d4, respectively. The team travels on two flights, and each flight should carry someone in every role, so that the team is complete even if the other flight is delayed.

Question 1.1.2. How can we assign the members to the 2 flights so that each flight has a person with every role?

The picture shows five members and the four underlined roles. Each closed curve encloses the members holding the role underlined in the same color, and the shading behind each member is a flight assignment: the members shaded red take one flight, those shaded green take the other. This answers the question, since each of the four curves contains both a red and a green member.

Abstractly, the data are a set of people with a family of subsets of it, one subset per role, and a flight assignment colors the people with two colors. This is the following structure.

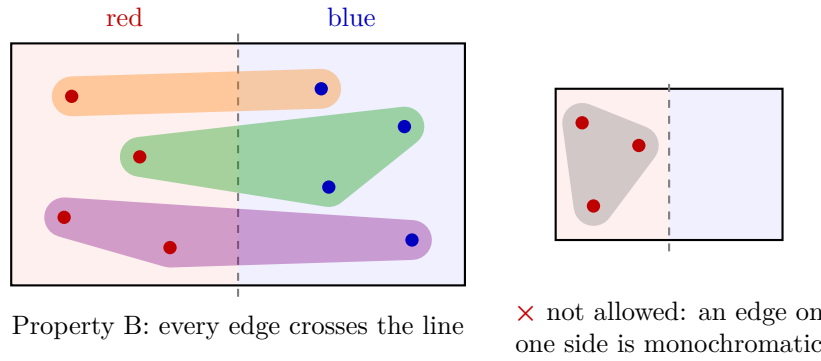
Definition 1.1.3 (Hypergraph). A **hypergraph** is a pair $H = (V, E)$ of vertices V and edges $E \subseteq 2^V$, where an edge is a subset of vertices. It is **k -uniform** if $|e| = k$ for all $e \in E$.

Notation . 2^V is the power set of V , and $V(H)$, $E(H)$ are the vertex and edge sets of H ; all hypergraphs here are finite. For a set X , $\binom{X}{k}$ is the family of all k -element subsets of X , and $[N] = \{1, \dots, N\}$.

Remark 1.1.4. A graph is a 2-uniform hypergraph: each of its edges is a set $\{u, v\}$ of two vertices.

Definition 1.1.5 (Property B). A hypergraph has **Property B**, or is **two-colorable**, if we can two-color the vertices of it s.t. every edge has vertices of both 2 colors.

Picture the two colors as the two halves of a box: put the red vertices in the left half and the blue vertices in the right half. Property B asks for a way to place every vertex so that each edge crosses the dividing line, i.e. has a vertex on both sides.



Equivalently, there is a coloring $\chi: V \rightarrow \{\text{red}, \text{blue}\}$ under which no edge is **monochromatic**, i.e. has all of its vertices of one color. The name refers to Felix Bernstein, who studied this property in 1908.

In [Example 1.1.1](#), the vertices are the members, each role is an edge, and a flight assignment is a two-coloring, so the question asks whether this hypergraph has Property B. The shading shows that the drawn part, a 3-uniform hypergraph with five vertices and four edges, does.

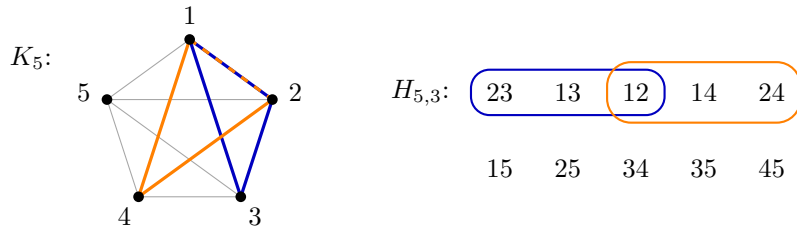
§1.1.ii. Connection to Ramsey Numbers

The Ramsey problem is a question about Property B for one particular family of hypergraphs. Given n and k , build the hypergraph $H_{n,k}$ with

$$V(H_{n,k}) = E(K_n), \quad E(H_{n,k}) = \{E(K) : K \text{ is a copy of } K_k \text{ in } K_n\}.$$

Its vertices are the $\binom{n}{2}$ edges of K_n . It has one edge for each k -set S of vertices of K_n , consisting of the $\binom{k}{2}$ edges of K_n inside S . Thus $H_{n,k}$ is a $\binom{k}{2}$ -uniform hypergraph with $\binom{n}{k}$ edges.

Example 1.1.6 ($n = 5, k = 3$). Write ij for the edge of K_5 between the vertices i and j .



The ten vertices of $H_{5,3}$ are the ten edges of K_5 , and its ten edges are the triangles of K_5 , each a set of three edges of K_5 . Only two triangles are drawn. The blue triangle 123 is the edge $\{12, 13, 23\}$ of $H_{5,3}$, and the orange triangle 124 is the edge $\{12, 14, 24\}$. These two edges of $H_{5,3}$ share the vertex 12 because the two triangles share the edge 12, drawn in both colors. The colors only mark the two triangles; they are not an edge-coloring.

A red/blue coloring of the edges of K_n is the same thing as a two-coloring of the vertices of $H_{n,k}$, and a copy of K_k is monochromatic exactly when the corresponding edge of $H_{n,k}$ is. Hence K_n has a coloring without a monochromatic K_k iff $H_{n,k}$ has Property B, and

$$R(k) = \min\{n : H_{n,k} \text{ does not have Property B}\}.$$

Unfolding the definition of Property B, $H_{n,k}$ is without Property B iff every two-coloring of its vertices leaves some edge monochromatic, so

$$R(k) = \min\{n : \forall \chi: V(H_{n,k}) \rightarrow \{\text{red}, \text{blue}\}, \exists e \in E(H_{n,k}) \text{ monochromatic under } \chi\}.$$

Since $V(H_{n,k}) = E(K_n)$, such a χ is a red/blue coloring of the edges of K_n , and a monochromatic $e \in E(H_{n,k})$ is a monochromatic copy of K_k ; this is word for word the definition of $R(k)$ from Lecture 1.

Remark 1.1.7. For 2-uniform hypergraphs, Property B $\equiv$ bipartite: a 2-uniform hypergraph is a graph, and an edge $\{u, v\}$ sees both colors iff u and v get different colors. So a two-coloring without monochromatic edges is exactly a partition of V into two independent sets, i.e. a bipartition. Property B thus generalizes bipartiteness to hypergraphs; for instance, C_5 from Lecture 1, and every other odd cycle, does not have Property B.

§1.1.iii. The Extremal Problem

Every edge is one more constraint on the coloring, so a hypergraph with few edges should be two-colorable. We measure the size of a hypergraph by its number of edges and leave the number of vertices unrestricted.

Question 1.1.8 (Extremal problem). How small can a k -uniform hypergraph be if it does not have Property B?

Definition 1.1.9.

$$m(k) := \min\{m : \exists k\text{-uniform hypergraph } H = (V, E), |E| = m, \text{ without Property B}\}.$$

Unfolding the definition of Property B, H is without Property B iff every two-coloring of V leaves some edge monochromatic, so

$$m(k) = \min \left\{ m : \begin{array}{l} \exists H = (V, E) \text{ with } |E| = m \text{ and } |e| = k \ \forall e \in E, \text{ s.t.} \\ \forall \chi: V \rightarrow \{\text{red, blue}\}, \exists e \in E \text{ monochromatic under } \chi \end{array} \right\}.$$

Here $\min \emptyset = \infty$, so $m(k) = \infty$ would mean that every k -uniform hypergraph has Property B; [Corollary 1.1.13](#) shows this does not happen. By minimality, every k -uniform hypergraph with fewer than $m(k)$ edges has Property B, while some k -uniform hypergraph with $m(k)$ edges does not.

Proposition 1.1.10 (Erdős, 1963)

$$m(k) \geq 2^{k-1}.$$

Proof. WTS: If $H = (V, E)$ is a k -uniform hypergraph with $m < 2^{k-1}$ edges, then it is two-colorable.

Color the vertices red and blue independently and uniformly. Let $E = \{e_1, e_2, \dots, e_m\}$, and let A_i be the event that the edge e_i is monochromatic. The edge e_i has exactly k vertices with independent colors, and it is monochromatic iff they are all red or all blue, so

$$\mathbb{P}(A_i) = 2 \left(\frac{1}{2} \right)^k = \left(\frac{1}{2} \right)^{k-1}.$$

The events A_i need not be independent, since edges may share vertices, but the union bound does not need independence:

$$\mathbb{P}(\exists \text{ monochromatic edge}) = \mathbb{P}\left(\bigcup_{i=1}^m A_i\right) \leq \sum_{i=1}^m \mathbb{P}(A_i) = m \cdot 2^{-(k-1)} < 1 \quad \text{for } m < 2^{k-1}.$$

Taking complements, and using that an event of positive probability contains at least one

outcome,

$$\begin{aligned}
 & \mathbb{P}(\# \text{ monochromatic edge}) > 0 \\
 & \Downarrow \\
 & \exists \text{ a two-coloring of vertices where every edge sees both colors} \\
 & \Downarrow \\
 & H \text{ is two-colorable.} \quad \blacksquare
 \end{aligned}$$

Remark 1.1.11. Applied to $H_{n,k}$, which is $\binom{k}{2}$ -uniform with $\binom{n}{k}$ edges, [Proposition 1.1.10](#) says: if $\binom{n}{k} < 2^{\binom{k}{2}-1}$, then $H_{n,k}$ has Property B, i.e. $R(k) > n$. This is exactly the condition $\binom{n}{k} 2^{1-\binom{k}{2}} < 1$ in Erdős's lower bound on $R(k)$, and the proof above is the same computation with K_n replaced by an arbitrary hypergraph.

§1.1.iv. Upper Bounds on $m(k)$

An upper bound on $m(k)$ comes from exhibiting one k -uniform hypergraph without Property B that has few edges.

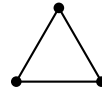
Proposition 1.1.12

$m(1) = 1$, $m(2) = 3$, and for every $k \geq 1$,

$$m(k) \leq \binom{2k-1}{k} \sim c \frac{2^{2k}}{\sqrt{k}}, \quad c = \frac{1}{2\sqrt{\pi}}.$$



$k = 1$: one vertex, one edge



$k = 2$: the triangle K_3

Proof.

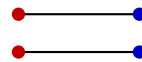
- (1) $m(1) = 1$: Take one vertex v and the single edge $\{v\}$ (left picture above). This edge has only one vertex, so it is monochromatic under every coloring. A hypergraph with no edges has Property B vacuously, so $m(1) = 1$. In [Example 1.1.1](#), an edge of size 1 is a role held by a single member, e.g. a lone sport psychologist: then no flight assignment works.
- (2) $m(2) = 3$: A 2-uniform hypergraph is a graph, and by [Remark 1.1.7](#) it is without Property B iff it is not bipartite. So $m(2)$ is the fewest edges of a non-bipartite graph, and we show two things.
 - $m(2) \leq 3$: The triangle K_3 (right picture above) is without Property B. Take any two-coloring of its three vertices. By the pigeonhole principle, two of the three vertices get the same color, and since every two vertices of K_3 are adjacent, the edge between them is monochromatic.
 - $m(2) \geq 3$: Every graph G with at most two edges has Property B. We color the vertices of the edges red and blue as below, and all other vertices, which lie in no edge, arbitrarily.

$\emptyset$

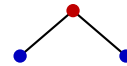
no edges



one edge



two disjoint
edges

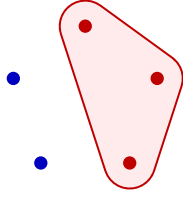


two edges sharing
a vertex

With one edge $\{u, v\}$, color u red and v blue. With two disjoint edges $\{a, b\}$, $\{c, d\}$, color a, c red and b, d blue. With two edges $\{a, b\}$, $\{a, c\}$ sharing the vertex a , color a red and b, c blue. In every case each edge has one red and one blue endpoint, so no edge is monochromatic.

Hence $m(2) = 3$. Note that [Proposition 1.1.10](#) alone only gives $m(2) \geq 2^{2-1} = 2$, so the case check above is needed for the lower bound.

- (3) $m(k) \leq \binom{2k-1}{k}$: Consider $2k - 1$ vertices and let any k of them form an edge, i.e. take the hypergraph $([2k - 1], \binom{[2k-1]}{k})$. In a two-coloring of $2k - 1$ vertices, by the pigeonhole principle, some color is used on at least k vertices, and any k of these form a monochromatic edge.



$k = 3$: five vertices, any three of which form an edge. In this coloring three vertices are red, and they form a monochromatic edge.

So this hypergraph does not have Property B, and $m(k) \leq \binom{2k-1}{k}$. For $k = 1, 2$, it is exactly the edge $\{v\}$ and the triangle above, so the bound is tight there.

- (4) Asymptotics: By Pascal's rule and $\binom{2k-1}{k-1} = \binom{2k-1}{k}$,

$$\binom{2k}{k} = \binom{2k-1}{k-1} + \binom{2k-1}{k} = 2\binom{2k-1}{k},$$

and Stirling's formula $N! \sim \sqrt{2\pi N}(N/e)^N$ gives

$$\binom{2k}{k} = \frac{(2k)!}{(k!)^2} \sim \frac{\sqrt{4\pi k} (2k/e)^{2k}}{2\pi k (k/e)^{2k}} = \frac{4^k}{\sqrt{\pi k}}.$$

Hence $\binom{2k-1}{k} = \frac{1}{2} \binom{2k}{k} \sim \frac{1}{2\sqrt{\pi}} \cdot \frac{2^{2k}}{\sqrt{k}}$. ■

Corollary 1.1.13

$m(k)$ is finite for every $k \geq 1$.

Proof. The hypergraph $([2k - 1], \binom{[2k-1]}{k})$ in the proof of [Proposition 1.1.12](#) is k -uniform and does not have Property B, so $m(k) \leq \binom{2k-1}{k} < \infty$. ■

Remark 1.1.14. Together, [Proposition 1.1.10](#) and [Proposition 1.1.12](#) give

$$2^{k-1} \leq m(k) \leq (1 + o(1)) \frac{4^k}{2\sqrt{\pi k}}.$$

Both bounds are exponential in k , with bases 2 and 4, the same kind of gap as $\sqrt{2}^k \lesssim R(k) \leq 4^k$ for Ramsey numbers. Unlike for Ramsey numbers, this gap can be closed up to a polynomial factor: the base 2 of the lower bound is the right one. The explicit hypergraph $([2k-1], \binom{[2k-1]}{k})$ is wasteful because it uses every k -set of its vertices, and the next subsection shows that a *random* choice of far fewer k -sets on a larger vertex set already fails to have Property B.

§1.1.v. A Random Construction

To show that $m(k) \leq m$ we must exhibit a k -uniform hypergraph with m edges and without Property B. Instead of constructing one, we pick one at random and show that it fails to have Property B with positive probability.

Theorem 1.1.15 (Erdős, 1964)

$m(k) = O(k^2 2^k)$, i.e. there is a constant $C > 0$ s.t. $m(k) \leq Ck^2 2^k$ for all large enough k .

Proof. Idea: take a random hypergraph H .

- Fix a set V of n vertices.
- Choose m edges $e_1, e_2, \dots, e_m$ uniformly at random from $\binom{V}{k}$, independently of each other.

The numbers n and m will be chosen later. The same k -set may be picked twice; then H has fewer than m distinct edges, and deleting repeated copies of an edge does not affect Property B. So it suffices to show

$$\mathbb{P}(H \text{ is not two-colorable}) > 0 \Rightarrow m(k) \leq |E(H)| \leq m.$$

Now the random object is the hypergraph, and the bad event is that H is two-colorable, i.e. for some two-coloring $\chi: V \rightarrow \{\text{red}, \text{blue}\}$ there is no monochromatic edge. There are 2^n colorings, so we fix one, bound the probability that it works, and take a union bound over all of them. This is the reverse of Proposition 1.1.10, where the hypergraph was fixed and the coloring was random.

Let χ be a two-coloring of V , and let A_χ be the event that none of $e_1, e_2, \dots, e_m$ is monochromatic under χ . Consider one edge e_i . Since e_i is uniformly random,

$$\mathbb{P}(e_i \text{ is monochromatic under } \chi) = \frac{\# \text{ monochromatic } k\text{-sets}}{\# \text{ possible } k\text{-sets}} = \frac{\binom{r}{k} + \binom{n-r}{k}}{\binom{n}{k}},$$

where r is the number of red vertices under χ . The function $x \mapsto \binom{x}{k} = \frac{x(x-1)\cdots(x-k+1)}{k!}$, extended by 0 for $x < k-1$, is convex on $[0, \infty)$.¹ By convexity,

$$\frac{1}{2} \left[\binom{r}{k} + \binom{n-r}{k} \right] \geq \binom{\frac{1}{2}(r+n-r)}{k} \Rightarrow \binom{r}{k} + \binom{n-r}{k} \geq 2 \binom{n/2}{k},$$

that is, a balanced coloring has the fewest monochromatic k -sets. Hence, for every χ ,

$$\mathbb{P}(e_i \text{ is monochromatic under } \chi) \geq \frac{2 \binom{n/2}{k}}{\binom{n}{k}} =: p,$$

where $\binom{n/2}{k}$ is the polynomial above evaluated at $x = n/2$, so n need not be even. The edges are chosen independently, so

$$\begin{aligned} \mathbb{P}(A_\chi) &= \mathbb{P}(\nexists \text{ monochromatic edge under } \chi) = \prod_{i=1}^m (1 - \mathbb{P}(e_i \text{ is monochromatic under } \chi)) \\ &\leq (1 - p)^m, \end{aligned}$$

¹On $[k-1, \infty)$ it is a polynomial all of whose roots are $\leq k-1$, so by Rolle's theorem all of its derivatives are positive there; gluing it to the constant 0 at $x = k-1$, where both vanish and the right derivative is ≥ 0 , keeps it convex.

and by the union bound over the 2^n colorings,

$$\mathbb{P}(H \text{ is two-colorable}) = \mathbb{P}\left(\bigcup_{\chi} A_{\chi}\right) \leq \sum_{\chi} \mathbb{P}(A_{\chi}) \leq 2^n(1-p)^m.$$

We must choose n and m so that $2^n(1-p)^m < 1$, where $p = 2^{\binom{n/2}{k}}/\binom{n}{k}$. Using $1-x < e^{-x}$ for $x > 0$,

$$2^n(1-p)^m < 2^n e^{-pm} = e^{n \ln 2 - pm}, \quad \text{where } e^{n \ln 2 - pm} \leq 1 \Leftrightarrow m \geq \frac{n \ln 2}{p}.$$

We will take $m = \lfloor \frac{n \ln 2}{p} \rfloor + 1$, and choose n to minimize it. For this we need p precisely as a function of n and k . Writing out both binomial coefficients, the $i = 0$ factor equals $\frac{1}{2}$ and cancels the 2:

$$\begin{aligned} p &= \frac{2^{\binom{n/2}{k}}}{\binom{n}{k}} = 2 \prod_{i=0}^{k-1} \frac{\frac{n}{2} - i}{n - i} = \prod_{i=1}^{k-1} \frac{n - 2i}{2(n - i)} = \prod_{i=1}^{k-1} \left(\frac{1}{2} - \frac{i}{2(n - i)} \right) = \frac{1}{2^{k-1}} \prod_{i=1}^{k-1} \left(1 - \frac{i}{n - i} \right) \\ &= \frac{1}{2^{k-1}} \prod_{i=1}^{k-1} \left(1 - \frac{i}{n} + O\left(\frac{i^2}{n^2}\right) \right) = \frac{1}{2^{k-1}} \prod_{i=1}^{k-1} e^{-\frac{i}{n} + O\left(\frac{i^2}{n^2}\right)} \\ &= \frac{1}{2^{k-1}} e^{-\sum_{i=1}^{k-1} \left(\frac{i}{n} + O\left(\frac{i^2}{n^2}\right) \right)} = \frac{1}{2^{k-1}} e^{-\frac{k(k-1)}{2n} + O\left(\frac{k^3}{n^2}\right)}. \end{aligned}$$

Here we assumed $n \geq 2k$, so that $\frac{i}{n-i} \leq 1$ and $\frac{i}{n-i} = \frac{i}{n} + O\left(\frac{i^2}{n^2}\right)$, and used $1-x = e^{-x+O(x^2)}$ for small $x \geq 0$. Now, since

$$m \approx \frac{n \ln 2}{p} = 2^{k-1} n e^{\frac{k(k-1)}{2n} + O\left(\frac{k^3}{n^2}\right)} \ln 2,$$

we minimize $n e^{k^2/(2n)}$ over n : its derivative in n is $e^{k^2/(2n)}(1 - \frac{k^2}{2n})$, which vanishes at $n = \frac{k^2}{2}$. The optimal choice is thus $n = ck^2$ with $c = \frac{1}{2}$, rounded to an integer. Then $\frac{k(k-1)}{2n} = 1 + O\left(\frac{1}{k}\right)$ and $\frac{k^3}{n^2} = O\left(\frac{1}{k}\right)$, so

$$m(k) \leq m \leq 2^{k-1} \cdot \frac{k^2}{2} \cdot e^{1+O(1/k)} \ln 2 + 1 = (1 + o(1)) \frac{e \ln 2}{4} k^2 2^k \leq C k^2 2^k. \quad \blacksquare$$

Remark 1.1.16 (Estimating binomial coefficients). Exact computation of p is not needed for $O(k^2 2^k)$, only for the constant. The standard estimates

$$\begin{aligned} \frac{(n-k)^k}{k!} &\leq \binom{n}{k} = \frac{n(n-1)(n-2) \cdots (n-k+1)}{k!} \leq \frac{n^k}{k!}, \\ \left(\frac{n}{k}\right)^k &\leq \binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k(k-1) \cdots (k-k+1)} \leq \left(\frac{ne}{k}\right)^k \end{aligned}$$

follow from bounding each factor $n-i$ between $n-k$ and n , from $\frac{n-i}{k-i} \geq \frac{n}{k}$, and from $k! \geq (k/e)^k$. The first one already gives

$$p = \frac{2^{\binom{n/2}{k}}}{\binom{n}{k}} \geq 2 \cdot \frac{(n/2 - k)^k / k!}{n^k / k!} = \frac{1}{2^{k-1}} \left(1 - \frac{2k}{n}\right)^k \geq \frac{1}{2^{k-1}} e^{-4k^2/n} \quad \text{for } n \geq 4k,$$

using $1-x \geq e^{-2x}$ for $0 \leq x \leq \frac{1}{2}$. With $n = 4k^2$ this yields $m(k) \leq (1 + o(1)) 2e \ln 2 \cdot k^2 2^k$, the same order with a worse constant.

§1.1.vi. Better Lower Bounds: Random Greedy Coloring

Combining [Proposition 1.1.10](#) and [Theorem 1.1.15](#),

Theorem 1.1.17 (Erdős, 1963–64)
$$2^{k-1} \leq m(k) \leq Ck^2 2^k \text{ for some constant } C > 0.$$

Since then the lower bound has been improved by polynomial factors in k , while the upper bound $O(k^2 2^k)$ is still the best known.

Theorem 1.1.18 (Radhakrishnan–Srinivasan, 2000)
$$m(k) \geq c \sqrt{\frac{k}{\ln k}} 2^k \text{ for some constant } c > 0.$$
Theorem 1.1.19 (Pluhár, 2009)
$$m(k) \geq c k^{1/4} 2^k \text{ for some constant } c > 0.$$

We now prove [Theorem 1.1.19](#). The proof has two ideas: first a deterministic greedy coloring, whose failures we can describe exactly, and then a random order of the vertices, which makes such failures unlikely.

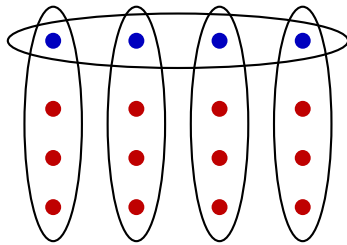
Proof. The proof of [Proposition 1.1.10](#) colored every vertex independently at random, which ignores the structure of H entirely. Instead:

Idea 1: Color the vertices greedily. Order the vertices $v_1, v_2, \dots, v_n$, and consider the vertices one by one. Color v_i red, unless that creates a monochromatic (red) edge, i.e. unless some edge contains v_i and all of its other vertices are already colored red, in which case we color v_i blue. Colors are never changed afterwards.

Observation .

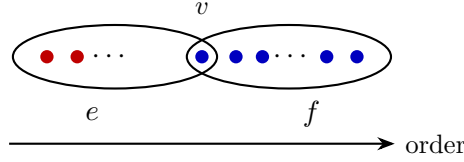
- (1) No monochromatic red edge: when the last vertex of an edge is reached and all the others are red, it is colored blue.
- (2) Monochromatic blue edges can occur. However, a vertex v is colored blue only because coloring it red would have completed a red edge. So every blue vertex v lies in some edge e such that all vertices of e other than v are red and come before v in the order.

For example, in the 4-uniform hypergraph below, order the twelve lower vertices first and the four top vertices last. The lower vertices are all colored red, since each vertical edge still has an uncolored top vertex. Each top vertex is then the last vertex of a vertical edge whose other vertices are red, so it is colored blue, and the horizontal edge ends up monochromatic blue.



Each blue vertex lies in a vertical edge that would otherwise be red.
The horizontal edge is monochromatic blue.

We say the greedy coloring *fails* if the coloring it produces is not a proper two-coloring, i.e. some edge is monochromatic. By (1), this happens exactly when some edge is monochromatic blue, as in the example. What must H and the order look like for this to happen? Suppose f is a monochromatic blue edge, and let v be the *first* vertex of f in the order. By (2), v lies in an edge e whose other vertices are red and come before v . Since all of f is blue, $e \cap f = \{v\}$; and since v is the last vertex of e and the first vertex of f , all of e comes before all of f :



We record what this argument shows. Note that it works for every order of the vertices.

Claim 1.1.1. For any order of the vertices, if the greedy coloring produces a monochromatic edge, then there are edges $e, f \in E(H)$ with $|e \cap f| = 1$ such that all of e comes before all of f in the order.

Here $e = f$ is possible only if $f = \{v\}$ is an edge of size 1: then $e = f = \{v\}$ and $|e \cap f| = 1$. Contrapositively, if no two edges meet in exactly one vertex, the greedy coloring cannot fail, whatever the order:

Corollary 1.1.20

If H is a hypergraph with $|e \cap f| \neq 1$ for all $e, f \in E(H)$, then H is two-colorable.

Proof. Fix any order $v_1, \dots, v_n$ of the vertices and run the greedy coloring. It colors every vertex red or blue, so it gives a two-coloring χ of V ; we show that no edge is monochromatic under χ .

- No edge is monochromatic red, by (1) of the observation.
- Suppose some edge f is monochromatic blue. By Claim 1.1.1 there would be edges e, f with $|e \cap f| = 1$, contradicting the assumption on H .

So every edge contains both colors, and H is two-colorable. ■

Idea 2: Order the vertices uniformly at random. Let H be a k -uniform hypergraph with m edges. Choose an order σ of V uniformly at random among all $n!$ orders, run the greedy coloring along σ , and let χ_σ be the resulting two-coloring. We will show that

$$\mathbb{P}(\chi_\sigma \text{ is not a proper two-coloring}) < 1 \quad \text{whenever } m < c k^{1/4} 2^k.$$

Then some order σ makes χ_σ proper, so H is two-colorable, and hence $m(k) \geq c k^{1/4} 2^k$.

Write $e \prec f$ if all of e comes before all of f in σ . By (1), χ_σ is not proper exactly when it has a monochromatic blue edge, and by Claim 1.1.1 this forces a pair $e \prec f$ with $|e \cap f| = 1$. Hence, by the union bound,

$$\begin{aligned} \mathbb{P}(\chi_\sigma \text{ is not proper}) &= \mathbb{P}(\exists \text{ monochromatic blue edge}) \\ &\leq \mathbb{P}(\exists e, f \in E(H) : |e \cap f| = 1, e \prec f) \\ &\leq \sum_{\substack{(e,f) \\ |e \cap f| = 1}} \mathbb{P}(e \prec f). \end{aligned}$$

Fix an ordered pair (e, f) with $|e \cap f| = 1$, so $|e \cup f| = 2k - 1$. Since σ is uniform, the order it induces on $e \cup f$ is uniform among all $(2k - 1)!$ orders of $e \cup f$. In an order with $e \prec f$, the $k - 1$ vertices of $e \setminus f$ come first, in any of $(k - 1)!$ orders, then the common vertex, then the

$k - 1$ vertices of $f \setminus e$, in any of $(k - 1)!$ orders. Thus

$$\mathbb{P}(e \prec f) = \frac{\overbrace{(k-1)!}^{\text{ordering } e \setminus f \text{ at front}} \overbrace{(k-1)!}^{\text{ordering } f \setminus e \text{ at back}}}{\underbrace{(2k-1)!}_{\text{total \# of orderings of } e \cup f}}.$$

There are at most m^2 ordered pairs (e, f) , so

$$\mathbb{P}(\chi_\sigma \text{ is not proper}) \leq m^2 \frac{(k-1)!(k-1)!}{(2k-1)!}.$$

By Stirling's approximation $N! \sim \sqrt{2\pi N}(N/e)^N$, in the form $\binom{2N}{N} \sim \frac{4^N}{\sqrt{\pi N}}$ from the proof of [Proposition 1.1.12](#),

$$m^2 \frac{(k-1)!(k-1)!}{(2k-1)!} = \frac{m^2}{(2k-1)\binom{2k-2}{k-1}} \sim \frac{m^2 \sqrt{\pi k}}{2k \cdot 4^{k-1}} = \frac{2\sqrt{\pi} m^2}{\sqrt{k} 2^{2k}},$$

which is < 1 if $m^2 < c'\sqrt{k} 2^{2k}$, i.e. if $m < ck^{1/4} 2^k$ (any $c < (2\sqrt{\pi})^{-1/2}$ works for large k). This is what we wanted to show. ■

Theorem 1.1.21 (Cherkashin–Kozik, 2014)

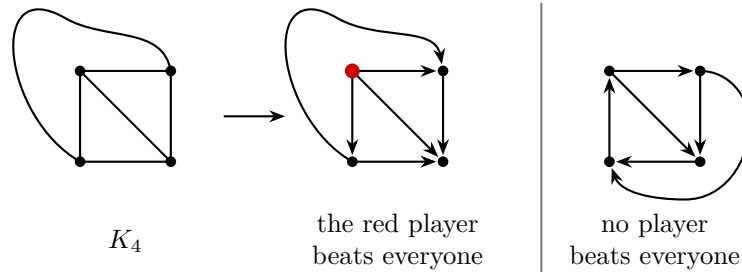
$$m(k) \geq c\sqrt{\frac{k}{\ln k}} 2^k \text{ for some constant } c > 0.$$

Proof idea. Exactly the same random greedy coloring of Pluhár, analyzed a little better. This recovers the bound of Radhakrishnan–Srinivasan with a much simpler proof. ■

§1.2. Schütte Tournaments

Definition 1.2.1 (Tournament). A **tournament** is an orientation of the edges of the complete graph K_n . **Orientation** means: for each edge $\{u, v\}$, we choose to orient the edge either from u to v , giving the arc (u, v) , or from v to u , giving the arc (v, u) . We say u **beats** v if $(u, v) \in E(T)$.

Think of a round-robin tournament: every two of the n players play one game, and the arc points from the winner to the loser. Sometimes the results determine a winner, a player who beats everyone else, and sometimes they do not.

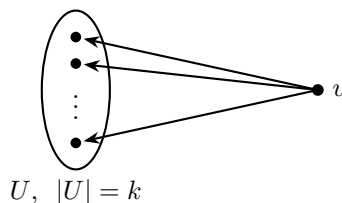


In the right tournament, the top-left player beats two others but loses to the bottom-left one, who in turn loses to both right players; every player loses to someone, so this condition cannot decide a winner. A winner is a single vertex u s.t. every other vertex is beaten by u . When there is no winner, we can relax the requirement: instead of one player who beats everyone, look for a small group of k players who together beat everyone else. This leads to the following question.

Question 1.2.2. Given a tournament T on n vertices, can we find a set U of k vertices s.t. for every vertex $v \notin U$, $\exists u \in U$ that beats v , i.e. $(u, v) \in E(T)$?

For $k = 1$ this asks for a winner. Schütte asked whether for every k there are tournaments in which no k players jointly beat everyone else.

Definition 1.2.3 (Schütte property). We say a tournament has the **k -Schütte property** if $\nexists$ such a set of k vertices, i.e. for any set U of k vertices, $\exists v \notin U$ s.t. v beats every $u \in U$.



Example 1.2.4. For $k = 1$, the 1-Schütte property says that there is no winner: every vertex is beaten by some other vertex. The right tournament above has it, and so does the cyclic triangle $a \rightarrow b \rightarrow c \rightarrow a$. No tournament on $n \leq 2$ vertices has it, so 3 is the fewest vertices for $k = 1$.

Question 1.2.5. How small can a tournament with the k -Schütte property be?

Lecture 3

Recall . A tournament T is an orientation of K_n , and it has the k -Schütte property if no k of its vertices jointly beat everyone else: for every $U \subseteq V(T)$ with $|U| = k$ there is a vertex $v \notin U$ beating every $u \in U$. For $k = 1$ this says that there is no winner.

29 Sep.

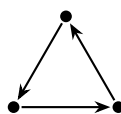
Section 1.2 ended by asking how small a tournament with the k -Schütte property can be. This lecture answers that question up to a polynomial factor, by the same pair of techniques used for $m(k)$ in **Section 1.1**: a deterministic greedy argument for the lower bound, and a random construction with a union bound for the upper bound.

§1.2.i. The Extremal Function

Definition 1.2.1. Let $f_S(k)$ be the smallest n s.t. there is an n -vertex tournament with the k -Schütte property.²

Since adding a vertex to a tournament with the k -Schütte property preserves the property (an old witness v for an old k -set U still works), tournaments with the k -Schütte property exist on every $n \geq f_S(k)$, and $f_S(k)$ is the threshold above which they appear.

Example 1.2.2. $f_S(1) = 3$. No tournament on 1 or 2 vertices has the 1-Schütte property: on one vertex v , the set $U = \{v\}$ has no vertex outside it; on two vertices, one of them beats the other and is therefore a winner. The cyclic triangle has it, since each of its three vertices is beaten by another one.



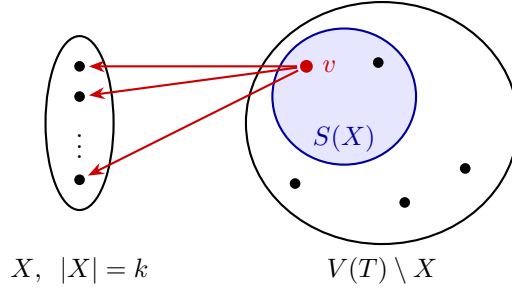
²Here $n > k$ is understood: if $n < k$ there is no k -set of vertices at all and the condition holds vacuously, and if $n = k$ the only k -set is $V(T)$, which has no vertex outside it.

To work with the k -Schütte property we name the set of vertices that a given set fails to dominate.

Notation . Given a tournament T and a subset $X \subseteq V(T)$ of its vertices, let

$$S(X) := \{v \in V(T) : (v, x) \in E(T) \ \forall x \in X\},$$

i.e. $S(X)$ is the set of vertices that beat everything in X .



A vertex v lies in $S(X)$ exactly when all k arcs between v and X point away from v , as drawn in red. Note that $S(\emptyset) = V(T)$, that S is order-reversing, i.e. $X \subseteq Y \Rightarrow S(Y) \subseteq S(X)$, and that $S(X) \cap X = \emptyset$ for $X \neq \emptyset$, since no vertex beats itself. The k -Schütte property is now a statement about S .

Lemma 1.2.3

T has the k -Schütte property $\Leftrightarrow S(X) \neq \emptyset$ for all $X \subseteq V(T)$ with $|X| = k$.

Both bounds below are proved through this reformulation. In particular, to show $f_S(k) > n$ we must rule out all n -vertex tournaments at once:

$$\begin{aligned} f_S(k) > n &\Leftrightarrow \text{no } n\text{-vertex tournament has the } k\text{-Schütte property} \\ &\Leftrightarrow \text{every } n\text{-vertex tournament has a set } X \text{ of size } k \text{ with } S(X) = \emptyset. \end{aligned}$$

So for the lower bound we are handed an arbitrary tournament and must *find* a bad k -set in it, which is what the next subsection does greedily.

§1.2.ii. Lower Bound: A Greedy Argument

The whole argument rests on the following averaging observation.

Lemma 1.2.4

In any tournament, there is a vertex that wins at least half of its games.

Proof. Count the total number of wins in a tournament T on n vertices. Every game produces exactly one winner, and there are $\binom{n}{2}$ games, so

$$\sum_{v \in V(T)} \#\{\text{wins of } v\} = \binom{n}{2}.$$

Since there are n vertices, the average number of wins per vertex is

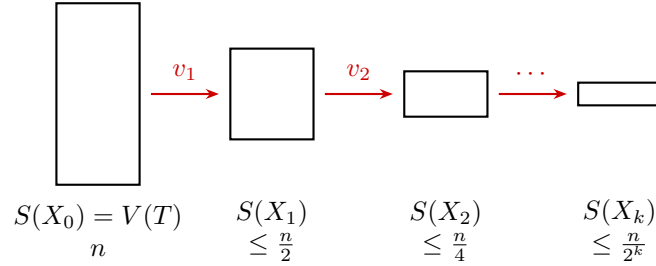
$$\frac{\binom{n}{2}}{n} = \frac{n-1}{2},$$

and some vertex is at least average, i.e. there is a vertex winning at least $\frac{n-1}{2}$ of the $n-1$ games it plays. ■

Proposition 1.2.5

$$f_S(k) \geq 2^k.$$

The idea is to build a bad k -set X one vertex at a time, choosing at each step a vertex that halves $S(X)$. Since $S(\emptyset) = V(T)$ has n elements, after k steps at most $n/2^k$ vertices are left, and if $n < 2^k$ there are none.



Proof. Let T be any tournament on $n < 2^k$ vertices; by Lemma 1.2.3 it suffices to produce a set X of k vertices with $S(X) = \emptyset$. Start with $X_0 = \emptyset$, so that $S(X_0) = V(T)$, and repeat the following for $i = 1, 2, \dots, k$.

If $S(X_{i-1}) = \emptyset$ already, stop. Otherwise apply Lemma 1.2.4 to the tournament $T[S(X_{i-1})]$ induced by T on $S(X_{i-1})$:

- find a vertex $v_i \in S(X_{i-1})$ beating at least half of the other vertices of $S(X_{i-1})$;
- add it to the set, $X_i := X_{i-1} \cup \{v_i\}$;
- this shrinks S to those vertices of $S(X_{i-1})$ that beat v_i , since

$$S(X_i) = S(X_{i-1}) \cap \{u : u \text{ beats } v_i\}.$$

Writing $s = |S(X_{i-1})|$, the vertex v_i beats at least $\frac{s-1}{2}$ of the remaining $s-1$ vertices, so at most $\frac{s-1}{2}$ of them beat v_i , and $v_i \notin S(X_i)$. Hence $S(X)$ shrinks by a factor of at least $\frac{1}{2}$ every round:

$$|S(X_i)| \leq \frac{|S(X_{i-1})| - 1}{2} \leq \frac{|S(X_{i-1})|}{2}.$$

After k rounds, $|X_k| = k$ and

$$|S(X_k)| \leq \frac{|V(T)|}{2^k} = \frac{n}{2^k} < 1,$$

so $S(X_k) = \emptyset$. If instead the process stopped early, at some X_i with $i < k$ and $S(X_i) = \emptyset$, extend X_i to any k -set $X \supseteq X_i$; as S is order-reversing, $S(X) \subseteq S(X_i) = \emptyset$. Either way T fails the k -Schütte property, so $f_S(k) \geq 2^k$. ■

§1.2.iii. Upper Bound: A Random Tournament

For the upper bound we must exhibit one tournament with the k -Schütte property. As in Theorem 1.1.15, we do not construct it: we choose a tournament at random and show that it has the property with positive probability.

Theorem 1.2.6 (Erdős)

If $\binom{n}{k} (1 - 2^{-k})^{n-k} < 1$, then $f_S(k) \leq n$.

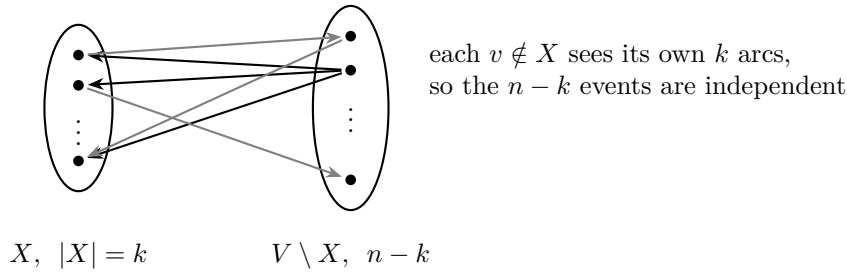
Proof. We consider a uniformly random tournament on the vertices $[n] = \{1, 2, \dots, n\}$, i.e. for every pair $\{i, j\}$ we choose the arc (i, j) or (j, i) uniformly and independently of all other pairs. By [Lemma 1.2.3](#), the bad events are indexed by the k -sets X , the event for X being that $S(X) = \emptyset$; we bound each one and take a union bound.

Fix X with $|X| = k$ and a vertex $v \in V \setminus X$. The vertex v lies in $S(X)$ iff all k arcs between v and X point away from v , and these k coin flips are independent, so

$$\mathbb{P}(v \in S(X)) = \mathbb{P}(v \text{ beats all of } X) = \left(\frac{1}{2}\right)^k \Rightarrow \mathbb{P}(v \notin S(X)) = 1 - \frac{1}{2^k}.$$

For distinct $v \notin X$, the events $\{v \in S(X)\}$ involve disjoint sets of pairs, namely the k pairs joining v to X , hence they are independent. Since $|V \setminus X| = n - k$,

$$\mathbb{P}(S(X) = \emptyset) = \mathbb{P}(\forall v \notin X, v \notin S(X)) = \prod_{v \notin X} \mathbb{P}(v \notin S(X)) = \prod_{v \notin X} \left(1 - \frac{1}{2^k}\right) = \left(1 - \frac{1}{2^k}\right)^{n-k}.$$



Now take the union bound over all $\binom{n}{k}$ choices of X :

$$\begin{aligned} \mathbb{P}(T \text{ is not } k\text{-Schütte}) &= \mathbb{P}(\exists X : S(X) = \emptyset) = \mathbb{P}\left(\bigcup_{X \in \binom{V}{k}} \{S(X) = \emptyset\}\right) \\ &\leq \sum_{X \in \binom{V}{k}} \mathbb{P}(S(X) = \emptyset) = \binom{n}{k} \left(1 - \frac{1}{2^k}\right)^{n-k} < 1 \end{aligned}$$

by assumption. Therefore $\mathbb{P}(T \text{ is } k\text{-Schütte}) > 0$, so there is an n -vertex tournament with the k -Schütte property, and $f_S(k) \leq n$. ■

It remains to find how small an n satisfies the hypothesis of [Theorem 1.2.6](#).

Corollary 1.2.7

$f_S(k) = O(k^2 2^k)$.

Proof. We estimate the two factors by

$$\binom{n}{k} \leq n^k, \quad (1 - 2^{-k})^{n-k} \leq e^{-(n-k)2^{-k}},$$

the second from $1 - x \leq e^{-x}$. So it suffices to have

$$n^k e^{-(n-k)2^{-k}} < 1 \Leftrightarrow k \ln n - (n-k)2^{-k} < 0 \Leftrightarrow n > k2^k \ln n + k.$$

The right-hand side grows only logarithmically in n , so the inequality holds as soon as n is a little larger than $k2^k \ln n$. Taking $n = \lceil k^2 2^k \rceil$ gives $\ln n = k \ln 2 + 2 \ln k + O(1) = (1 + o(1))k \ln 2$, hence

$$k2^k \ln n + k = (1 + o(1))(\ln 2)k^2 2^k < k^2 2^k \leq n$$

for large k , as $\ln 2 < 1$. Thus $n = \Theta(k^2 2^k)$ suffices, and $f_S(k) = O(k^2 2^k)$. $\blacksquare$

Putting [Proposition 1.2.5](#) and [Corollary 1.2.7](#) together:

Theorem 1.2.8

$2^k \leq f_S(k) \leq Ck^2 2^k$ for some constant $C > 0$.

Remark 1.2.9. This is the same picture as for Property B: an easy greedy lower bound 2^k , a random upper bound $O(k^2 2^k)$, and a gap of a polynomial factor $O(k^2)$ between them. Compare $2^{k-1} \leq m(k) \leq Ck^2 2^k$ from [Section 1.1](#), where the lower bound has since been improved to $c\sqrt{k/\ln k} 2^k$.

Exercise 1.2.10. Show that $f_S(k) \geq ck^2 2^k$ for some constant $c > 0$, improving [Proposition 1.2.5](#) by a factor of k .

§1.3. The Erdős–Ko–Rado Theorem

§1.3.i. Intersecting Families

Property B asked how to *split* a family of sets; we now ask how large a family can be if no two of its members can be split apart at all, i.e. if they pairwise overlap.

Definition 1.3.1 (Intersecting family). A family $\mathcal{F} \subseteq 2^{[n]}$ is **intersecting** if $A \cap B \neq \emptyset$ for any two members $A, B \in \mathcal{F}$.

Question 1.3.2. How large can an intersecting family be?

There are two natural ways to build a large one: take only *big* sets, which are forced to overlap for lack of room, or take only sets through one fixed element.

Example 1.3.3 (Large sets). For $n \geq 3$, the family $\mathcal{F} = \{[n]\} \cup \{[n] \setminus \{x\} : x \in [n]\}$ is intersecting, since any two of its members miss at most two elements between them. More generally, take every set of size more than $\frac{n}{2}$,

$$\mathcal{F} = \binom{[n]}{n} \cup \binom{[n]}{n-1} \cup \dots \cup \binom{[n]}{\lceil \frac{n+1}{2} \rceil}.$$

If $|A|, |B| > \frac{n}{2}$ then $|A| + |B| > n$, so there is not enough space in $[n]$ for A and B to be disjoint. For odd n , exactly half of all subsets have size more than $\frac{n}{2}$, so $|\mathcal{F}| = 2^{n-1}$.

Example 1.3.4 (Stars). Fix $x \in [n]$ and let $\mathcal{F}_x = \{F \subseteq [n] : x \in F\}$ be the **star** at x , i.e. the family of *all* subsets containing x . Any two members share x , so $\mathcal{F}_x$ is intersecting, and $|\mathcal{F}_x| = 2^{n-1}$, since a subset containing x is determined by its intersection with $[n] \setminus \{x\}$.

Both examples reach 2^{n-1} , and nothing does better.

Proposition 1.3.5

If $\mathcal{F} \subseteq 2^{[n]}$ is intersecting, then $|\mathcal{F}| \leq 2^{n-1}$.

Proof. Partition all 2^n subsets into the 2^{n-1} complementary pairs $\{F, [n] \setminus F\}$. The two members of a pair are disjoint, so an intersecting family contains at most one set from each pair, whence $|\mathcal{F}| \leq 2^{n-1}$. ■

Example 1.3.3 is really a cheat: it wins by using only large sets. The question becomes interesting once all sets are required to have the same size.

Question 1.3.6 (Erdős–Ko–Rado). What is the largest intersecting family $\mathcal{F} \subseteq \binom{[n]}{k}$?

If $n < 2k$, then any two k -sets $A, B \subseteq [n]$ satisfy $|A| + |B| = 2k > n$ and hence intersect, so $\binom{[n]}{k}$ is itself intersecting and the answer is $\binom{[n]}{k}$. **From now on we assume** $n \geq 2k$. The star of **Example 1.3.4**, cut down to k -sets, still works.

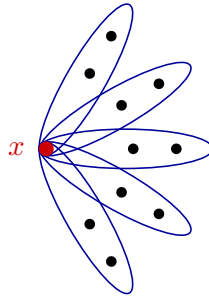
Example 1.3.7 (The k -uniform star). Fix $x \in [n]$. The **star** at x is

$$\mathcal{F}_x^{(k)} := \left\{ F \in \binom{[n]}{k} : x \in F \right\},$$

the family consisting of *every* k -set that contains x , and of nothing else. It is intersecting, as any two of its members share x . Choosing a k -set through x amounts to choosing its other $k-1$ elements from $[n] \setminus \{x\}$, so

$$|\mathcal{F}_x^{(k)}| = \binom{n-1}{k-1}, \quad \frac{|\mathcal{F}_x^{(k)}|}{|\binom{[n]}{k}|} = \frac{k}{n}.$$

In the picture, each ellipse is one member of the star, and they meet exactly in the red element x .



More generally, call a family $\mathcal{F} \subseteq \binom{[n]}{k}$ **trivially intersecting** if all of its members share a common element, i.e. $\bigcap_{F \in \mathcal{F}} F \neq \emptyset$. Such a family is certainly intersecting. The two notions are related but not identical: $\mathcal{F}$ is trivially intersecting exactly when $\mathcal{F} \subseteq \mathcal{F}_x^{(k)}$ for some x , so the trivially intersecting families are precisely the subfamilies of stars, and the stars are the largest of them. In particular a trivially intersecting family of the full size $\binom{n-1}{k-1}$ is a star.

The Erdős–Ko–Rado theorem says that for $n \geq 2k$ a star is already optimal.

Theorem 1.3.8 (Erdős–Ko–Rado, 1961)

If $n \geq 2k$ and $\mathcal{F} \subseteq \binom{[n]}{k}$ is intersecting, then $|\mathcal{F}| \leq \binom{n-1}{k-1}$.

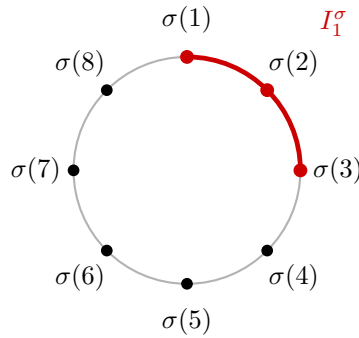
§1.3.ii. Katona's Cycle Method

The proof we give is Katona's, and it is a probabilistic argument of a shape we have not used yet. Rather than bounding $|\mathcal{F}|$ directly, we exhibit a random k -set A that is uniform on $\binom{[n]}{k}$, so that $\mathbb{P}(A \in \mathcal{F}) = |\mathcal{F}|/\binom{[n]}{k}$, and then bound that same probability by $\frac{k}{n}$ using a completely different description of A . Comparing the two expressions gives the theorem.

The second description arranges $[n]$ around a circle. Fix n positions $1, 2, \dots, n$ around a circle, and let a **cyclic permutation** σ of $[n]$ be a placement of the n elements at those positions, i.e. a bijection from positions to elements, with $\sigma(i)$ the element at position i . There are $n!$ of them. The word *cyclic* refers to the positions being read modulo n , so that position n is followed by position 1. Call

$$I_i^\sigma = \{\sigma(i), \sigma(i+1), \dots, \sigma(i+k-1)\}, \quad i \in [n],$$

with indices read modulo n , the **intervals of length k** of σ . There are exactly n of them, one per starting position, and most k -sets are not intervals of any given σ .



The point of intervals is that a circle has very little room for pairwise intersecting ones.

Lemma 1.3.9 (Katona)

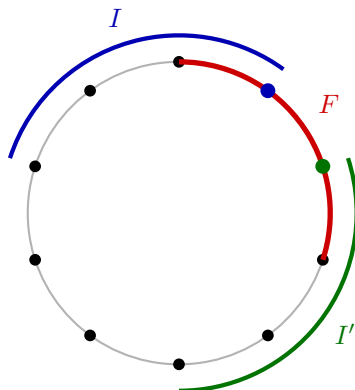
Let $n \geq 2k$ and let $\mathcal{F} \subseteq \binom{[n]}{k}$ be intersecting. Then for any cyclic permutation σ of $[n]$, at most k of the n intervals of length k of σ belong to $\mathcal{F}$.

Proof. Fix σ . If no interval lies in $\mathcal{F}$ we are done, so fix an interval $F \in \mathcal{F}$; after rotating the labels we may assume $F = \{\sigma(1), \dots, \sigma(k)\}$.

First we count the other intervals meeting F . An interval other than F meets F exactly when it starts at one of $\sigma(2), \dots, \sigma(k)$, or ends at one of $\sigma(1), \dots, \sigma(k-1)$. Since $n \geq 2k$, these $(k-1) + (k-1) = 2k-2$ intervals are distinct and none of them is F .

Now pair them up. For each i with $1 \leq i \leq k-1$, pair

$$\underbrace{\{\sigma(i-k+1), \dots, \sigma(i)\}}_{\text{the interval ending at } \sigma(i)} \quad \text{with} \quad \underbrace{\{\sigma(i+1), \dots, \sigma(i+k)\}}_{\text{the interval starting at } \sigma(i+1)}.$$



I ends at the blue vertex $\sigma(i)$,
 I' starts at the green vertex $\sigma(i+1)$;
together they span $2k \leq n$ positions,
so $I \cap I' = \emptyset$.

These two intervals occupy $2k$ consecutive positions of the circle, and $n \geq 2k$, so they are *disjoint*. Every one of the $2k - 2$ intervals meeting F occurs in exactly one such pair, so they split into $k - 1$ disjoint pairs.

Since $\mathcal{F}$ is intersecting it contains at most one interval from each disjoint pair, and every interval of $\mathcal{F}$ other than F must meet F . Hence $\mathcal{F}$ contains at most $(k - 1) + 1 = k$ intervals of σ . ■

Proof of Theorem 1.3.8. Define a random k -set $A \in \binom{[n]}{k}$ as follows.

- (1) Choose a uniformly random cyclic permutation σ of $[n]$.
- (2) Choose $i \in [n]$ uniformly at random, independently of σ , and set $A = I_i^\sigma = \{\sigma(i), \sigma(i + 1), \dots, \sigma(i + k - 1)\}$.

We first check that A is uniform on $\binom{[n]}{k}$. Fix $S \in \binom{[n]}{k}$ and count the pairs (σ, i) with $I_i^\sigma = S$: the start i is any of the n positions, and σ must then place S at the positions $i, \dots, i + k - 1$ and $[n] \setminus S$ at the rest, in $k! (n - k)!$ ways. So $n \cdot k! (n - k)!$ pairs produce S , a count not depending on S , and as (σ, i) is uniform among all $n \cdot n!$ pairs,

$$\mathbb{P}(A = S) = \frac{n \cdot k! (n - k)!}{n \cdot n!} = \frac{1}{\binom{n}{k}} \quad \text{for every } S.$$

Therefore

$$\mathbb{P}(A \in \mathcal{F}) = \frac{|\mathcal{F}|}{\left| \binom{[n]}{k} \right|} = \frac{|\mathcal{F}|}{\binom{n}{k}}. \quad (1)$$

On the other hand, condition on σ . As i was chosen independently of σ , it is still uniform on $[n]$, so A is one of the n intervals of σ , each with probability $\frac{1}{n}$, and

$$\mathbb{P}(A \in \mathcal{F} \mid \sigma) = \frac{\#\{i \in [n] : I_i^\sigma \in \mathcal{F}\}}{n} \leq \frac{k}{n} \quad \text{for every } \sigma.$$

The inequality is [Lemma 1.3.9](#): since $k < n$, distinct starts give distinct intervals, so the numerator counts the intervals of σ lying in $\mathcal{F}$. Averaging over σ ,

$$\mathbb{P}(A \in \mathcal{F}) = \sum_{\sigma} \mathbb{P}(\sigma) \underbrace{\mathbb{P}(A \in \mathcal{F} \mid \sigma)}_{\leq k/n} \leq \frac{k}{n} \sum_{\sigma} \mathbb{P}(\sigma) = \frac{k}{n}. \quad (2)$$

Comparing (1) and (2),

$$\frac{|\mathcal{F}|}{\binom{n}{k}} \leq \frac{k}{n} \Rightarrow |\mathcal{F}| \leq \frac{k}{n} \binom{n}{k} = \binom{n-1}{k-1}. \quad \text{■}$$

[Theorem 1.3.8](#) says that no intersecting family beats a star. In fact nothing else even ties with one.

Corollary 1.3.10 (Uniqueness in Erdős–Ko–Rado)

Let $n > 2k$ and let $\mathcal{F} \subseteq \binom{[n]}{k}$ be intersecting with $|\mathcal{F}| = \binom{n-1}{k-1}$. Then $\mathcal{F} = \mathcal{F}_x^{(k)}$ for some $x \in [n]$, i.e. the extremal families are exactly the stars.

Proof sketch. It is enough to show that $\mathcal{F}$ is trivially intersecting: then $\mathcal{F} \subseteq \mathcal{F}_x^{(k)}$ for some x , and as $|\mathcal{F}_x^{(k)}| = \binom{n-1}{k-1} = |\mathcal{F}|$ the two families coincide.

For $k = 1$ an intersecting family of singletons has a single member $\{x\}$, which is the star at x ; so assume $k \geq 2$ and suppose $\mathcal{F}$ were *not* trivially intersecting. [Theorem 1.3.12](#) below, whose proof is independent of this corollary, then gives

$$|\mathcal{F}| \leq \binom{n-1}{k-1} - \binom{n-k-1}{k-1} + 1.$$

Since $n > 2k$ we have $n - k - 1 \geq k$, so $\binom{n-k-1}{k-1} \geq \binom{k}{k-1} = k \geq 2$, and therefore $|\mathcal{F}| < \binom{n-1}{k-1}$, contradicting the hypothesis. ■

Remark 1.3.11. The strict inequality $n > 2k$ is needed. For $n = 2k$ the k -sets fall into $\frac{1}{2}\binom{2k}{k}$ complementary pairs $\{A, [n] \setminus A\}$, and any two non-complementary k -sets meet; so choosing one set out of each pair gives an intersecting family of size $\frac{1}{2}\binom{2k}{k} = \binom{2k-1}{k-1}$, which is extremal. Choosing inconsistently leaves it with no common element, so for $n = 2k$ the extremal families are far from unique.

If we forbid the star, the maximum drops sharply.

Theorem 1.3.12 (Hilton–Milner, 1967)

If $n > 2k$ and $\mathcal{F} \subseteq \binom{[n]}{k}$ is non-trivially intersecting, i.e. $\bigcap_{F \in \mathcal{F}} F = \emptyset$, then

$$|\mathcal{F}| \leq \binom{n-1}{k-1} - \binom{n-k-1}{k-1} + 1.$$

Remark 1.3.13. For $n \gg k^2$ we have $\binom{n-k-1}{k-1} \approx \binom{n-1}{k-1}$, so the Hilton–Milner bound is a vanishing fraction of the Erdős–Ko–Rado bound: once a common element is forbidden, an intersecting family must be much smaller.

§1.4. Bollobás Set-Pair Systems

Katona’s proof averaged over random orderings of a circle. The next theorem averages over random orderings of a line, and it is one of the most useful counting tools in extremal set theory.

Definition 1.4.1 (Set-pair system). A **Bollobás set-pair system** is a sequence of pairs of sets $\{(A_i, B_i) : 1 \leq i \leq m\}$ s.t.

- (1) $A_i \cap B_i = \emptyset$ for every i ;
- (2) $A_i \cap B_j \neq \emptyset$ for every $i \neq j$.

So each B_i is a set that A_i avoids but that every other A_j hits: B_i is a certificate distinguishing the index i from all the others. The theorem says that such certificates cannot be too numerous if they are small.

Theorem 1.4.2 (Bollobás, 1965)

Let $\{(A_i, B_i) : 1 \leq i \leq m\}$ be a set-pair system with $|A_i| = k$ and $|B_i| = \ell$ for all i . Then

$$m \leq \binom{k+\ell}{k}.$$

Proof. Let $X = \bigcup_i (A_i \cup B_i)$ be the ground set.

- (1) Take a uniformly random ordering of the elements of X .
- (2) For each i , define the event

$$E_i = \{\text{all elements of } A_i \text{ come before all elements of } B_i\}.$$

Since $A_i \cap B_i = \emptyset$, the set $A_i \cup B_i$ has exactly $k + \ell$ elements, and the random ordering restricted to it is uniform among its $(k + \ell)!$ orderings. Of these, $k! \ell!$ put all of A_i first, so

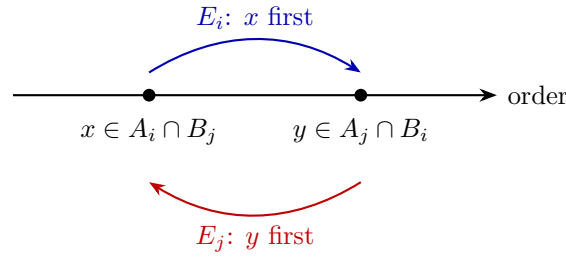
$$\mathbb{P}(E_i) = \frac{k! \ell!}{(k + \ell)!} = \frac{1}{\binom{k + \ell}{k}}.$$

Claim 1.4.1. The events $E_1, \dots, E_m$ are pairwise disjoint.

Proof. Suppose E_i and E_j both occur for some $i \neq j$. By condition (2) of Definition 1.4.1 applied both ways, we may pick

$$x \in A_i \cap B_j \quad \text{and} \quad y \in A_j \cap B_i.$$

Since E_i occurs and $x \in A_i$, $y \in B_i$, the element x comes before y . Since E_j occurs and $y \in A_j$, $x \in B_j$, the element y comes before x . This is a contradiction.



■

Being pairwise disjoint, the probabilities of the E_i add up, and they cannot exceed 1:

$$1 \geq \mathbb{P}\left(\bigcup_{i=1}^m E_i\right) = \sum_{i=1}^m \mathbb{P}(E_i) = \sum_{i=1}^m \frac{1}{\binom{k + \ell}{k}} = \frac{m}{\binom{k + \ell}{k}},$$

which is exactly $m \leq \binom{k + \ell}{k}$.

■

Remark 1.4.3 (The bound is attained). Take the ground set $[k + \ell]$, let $A_1, \dots, A_m$ list *all* of its k -element subsets, so that $m = \binom{k + \ell}{k}$, and let $B_i = [k + \ell] \setminus A_i$ be the complement of A_i . This is a set-pair system with the required sizes:

- $|A_i| = k$ and $|B_i| = (k + \ell) - k = \ell$;
- $A_i \cap B_i = \emptyset$, as B_i is the complement of A_i ;
- for $i \neq j$, $A_i \cap B_j = A_i \setminus A_j \neq \emptyset$, because A_i and A_j are distinct sets of the same size, so neither can contain the other.

So Theorem 1.4.2 is sharp for every k and ℓ .

§1.5. Application: Saturated Graphs

Throughout this section H is a fixed graph, thought of as a pattern we want to avoid.

Definition 1.5.1 (H -free). A **copy of H** in a graph G is a subgraph of G isomorphic to H , i.e. an injection $\varphi: V(H) \rightarrow V(G)$ with $\varphi(u)\varphi(v) \in E(G)$ for every edge $uv \in E(H)$. If G contains no copy of H , we call G **H -free**.

Remark 1.5.2. A copy need not be *induced*: G may have further edges among the vertices $\varphi(V(H))$, and they do not stop φ from being a copy. So G is K_k -free exactly when no k vertices of G are pairwise adjacent.

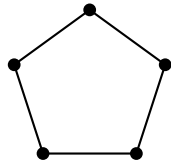
Classical extremal graph theory asks for the *maximum* number of edges of an H -free graph.

Question 1.5.3 (Mantel). How many edges can an n -vertex graph have if it does not contain a triangle?

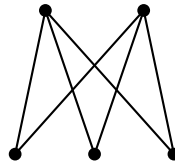
Question 1.5.4 (Turán, 1941). How many edges can an n -vertex graph have if it does not contain K_k ? Write $\text{ex}(n, H)$ for this maximum, the **Turán number** of H .

One might try to construct such a graph greedily: consider the edges of K_n one by one, and add an edge to our graph if it does not create a copy of K_k . This certainly produces a K_k -free graph, but it need not produce one with the maximum number of edges: the process only guarantees that no further edge *can* be added.

Example 1.5.5. Take $n = 5$ and $k = 3$. Both graphs below are triangle-free, and in both of them adding any missing edge creates a triangle, so either can be the output of the greedy algorithm. They have a different number of edges.



C_5 : 5 edges



$K_{2,3}$: 6 edges

So the greedy algorithm computes a *maximal* H -free graph rather than a maximum one, and it is natural to ask how few edges a maximal H -free graph can have.

Definition 1.5.6 (Saturated graph). Given a fixed graph H , a graph G is **H -saturated** if

- (1) G is H -free, and
- (2) adding any missing edge to G creates a copy of H ,

i.e. G is maximally H -free.

Observation . G is H -saturated $\Leftrightarrow G$ is the output of the H -free greedy algorithm for some ordering of the edges of K_n .

Proof. ($\Leftarrow$) The output G is H -free by construction. If f is a missing edge, it was rejected at its turn because adding it to the graph G' built so far created a copy of H . As $G' \subseteq G$, that copy is still present in $G + f$.

($\Rightarrow$) Order the edges of K_n so that those of G come first. Every prefix of $E(G)$ is a subgraph of the H -free graph G , so the algorithm accepts all of them. Afterwards the current graph is G , and each remaining edge f is rejected since $G + f$ contains a copy of H by saturation. The output is G . ■

Definition 1.5.7 (Saturation number). Given a graph H , the **saturation number** $\text{sat}(n, H)$ is the minimum number of edges in an n -vertex H -saturated graph.

Remark 1.5.8 (Why the minimum). The maximum gives nothing new: it is the Turán number $\text{ex}(n, H)$ of [Question 1.5.4](#) again. An H -saturated graph is H -free, so has at most $\text{ex}(n, H)$ edges; conversely an H -free graph with $\text{ex}(n, H)$ edges is saturated, as adding any edge leaves too many for an H -free graph. So $\text{sat}(n, H) \leq e(G) \leq \text{ex}(n, H)$, with both ends attained.

Notation (Subscripted O). A subscript on O records which parameters the implied constant is allowed to depend on. Thus $f(n) = O_k(n)$ means: for each fixed k there is a constant C_k , depending on k but *not* on n , with $|f(n)| \leq C_k n$ for all n . Plain $O(\cdot)$, as used earlier, allows no such dependence. Dually, $g(n) = \Omega(n^2)$ means $g(n) \geq cn^2$ for some constant $c > 0$ and all large n .

Theorem 1.5.9 (Bollobás, 1965)

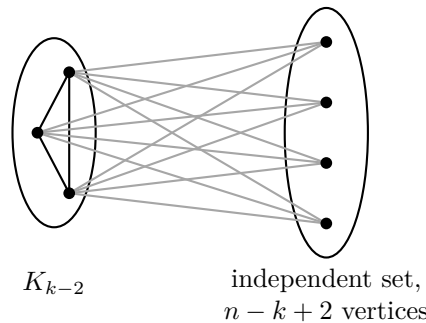
For every $k \geq 2$,

$$\text{sat}(n, K_k) = \binom{n}{2} - \binom{n-k+2}{2} = (k-2)n - \binom{k-1}{2} = O_k(n).$$

Here $\text{sat}(n, K_k) = (k-2)n - \binom{k-1}{2} \leq (k-2)n$, so $C_k = k-2$ works: for each fixed k the saturation number grows only linearly in n , though the slope $k-2$ grows with k .

Remark 1.5.10. Note how far this is from the Turán problem: for $k \geq 3$ the largest K_k -free graphs have $\Omega(n^2)$ edges, whereas the smallest maximal ones have only $O_k(n)$.

Remark 1.5.11 (The bound is best possible). Join a clique K_{k-2} to an independent set of $n - k + 2$ vertices, i.e. put in all edges inside the clique and all edges between the two parts, and none inside the independent set.



This graph is K_k -free, since a clique may use at most one vertex of the independent set and so has at most $k-1$ vertices; and adding any missing edge, which joins two vertices of the independent set, completes a K_k . Its number of edges is

$$\binom{k-2}{2} + (k-2)(n-k+2) = \binom{n}{2} - \binom{n-k+2}{2},$$

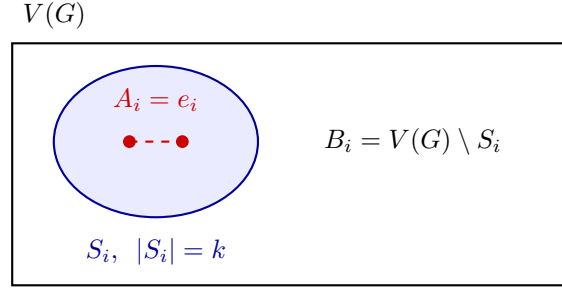
so $\text{sat}(n, K_k)$ is at most the claimed value.

It remains to prove the matching lower bound, and this is where set pairs enter: the missing edges of a saturated graph, together with the cliques they complete, form a set-pair system.

Proof of the lower bound in [Theorem 1.5.9](#). Let G be a smallest K_k -saturated graph on n vertices, and let $e_1, e_2, \dots, e_m$ be its missing edges, so that

$$m = \binom{n}{2} - e(G).$$

We construct a set-pair system. Put $A_i = e_i$, a set of two vertices. By saturation, adding e_i to G creates a k -clique, say on the vertex set S_i with $|S_i| = k$; this clique must use the new edge, since G itself is K_k -free, so $e_i \subseteq S_i$. Put $B_i = V(G) \setminus S_i$, a set of $n - k$ vertices.



We check the two conditions of [Definition 1.4.1](#).

- $A_i \cap B_i = e_i \cap (V(G) \setminus S_i) = \emptyset$, since $e_i \subseteq S_i$.
- Let $i \neq j$ and suppose $A_i \cap B_j = \emptyset$, i.e. $e_i \subseteq S_j$. Adding e_j to G makes S_j a clique, so every pair inside S_j lies in $E(G) \cup \{e_j\}$. But e_i is a pair inside S_j that is neither an edge of G nor equal to e_j , so it is still missing and S_j is not a clique in $G + e_j$. Contradiction; hence $A_i \cap B_j \neq \emptyset$.

Thus $\{(A_i, B_i) : 1 \leq i \leq m\}$ is a set-pair system with $|A_i| = 2$ and $|B_i| = n - k$ for all i , and [Theorem 1.4.2](#) gives

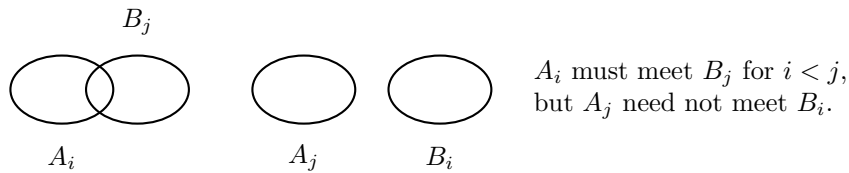
$$m \leq \binom{2 + (n - k)}{2} = \binom{n - k + 2}{2} \Rightarrow e(G) = \binom{n}{2} - m \geq \binom{n}{2} - \binom{n - k + 2}{2}. \quad \blacksquare$$

Remark 1.5.12.

- (1) The same argument works for complete hypergraphs, with A_i the missing edge and B_i the complement of the clique it completes.
- (2) **Weak saturation** asks instead that the missing edges can be added one by one *in some order*, each addition creating a new copy of H . Running the argument above only gives the weaker conditions

$$A_i \cap B_i = \emptyset, \quad A_i \cap B_j \neq \emptyset \text{ for } i < j,$$

i.e. condition (2) of [Definition 1.4.1](#) is required for one of the two orders only. It is still true that $m \leq \binom{k+\ell}{k}$ for such a *skew* set-pair system, but the random-ordering proof breaks down, because the events E_i are no longer pairwise disjoint, and the only known proofs are algebraic.



2. Linearity of Expectation

Lecture 4

So far every probabilistic argument had the same shape: list the bad events, bound the probability of each, and use the union bound to show that with positive probability none of them happens. This chapter replaces probabilities by *expectations*. The basic principle is that a random variable cannot always be above its average, and the tool that makes averages computable is linearity of expectation.

2 Oct.

§2.1. Expectation and the First Moment Principle

Throughout, $(\Omega, \mathbb{P})$ is a probability space and $X: \Omega \rightarrow \mathbb{R}$ is a random variable. In all our applications Ω is finite, and then the **expectation** of X is its weighted average value

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\omega).$$

§2.1.i. A Random Variable Is Not Always Above Its Mean

Lemma 2.1.1 (First moment principle)

The events $\{X \leq \mathbb{E}[X]\}$ and $\{X \geq \mathbb{E}[X]\}$ both have positive probability. In particular

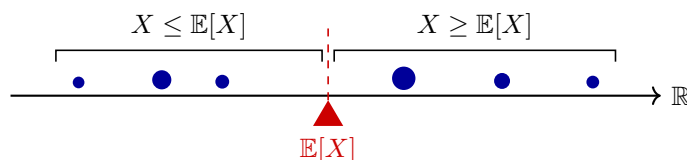
$$\exists \omega \in \Omega \text{ s.t. } X(\omega) \leq \mathbb{E}[X] \quad \text{and} \quad \exists \omega' \in \Omega \text{ s.t. } X(\omega') \geq \mathbb{E}[X].$$

Proof. Suppose $\mathbb{P}(X \geq \mathbb{E}[X]) = 0$. Then $X(\omega) < \mathbb{E}[X]$ for every ω with $\mathbb{P}(\omega) > 0$, and averaging over these ω gives

$$\mathbb{E}[X] = \sum_{\omega: \mathbb{P}(\omega) > 0} X(\omega) \mathbb{P}(\omega) < \sum_{\omega: \mathbb{P}(\omega) > 0} \mathbb{E}[X] \mathbb{P}(\omega) = \mathbb{E}[X],$$

a contradiction. The other event is handled in the same way. ■

Think of the values of X as weights on the real line: $\mathbb{E}[X]$ is the point where they balance, so there must be weight on each side of it, or on it.



This is how we will use it: to show that some object with a property exists, define a random object and a random variable X measuring the property, compute $\mathbb{E}[X]$, and conclude that some outcome does at least as well as the average.

§2.1.ii. Linearity

Computing $\mathbb{E}[X]$ directly is usually hopeless, since the distribution of X is complicated. The point is that we never need the distribution: we break X into simple pieces.

Theorem 2.1.2 (Linearity of expectation)

If $X = \sum_{i=1}^n \alpha_i X_i$, where the X_i are random variables and the α_i are constants, then

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^n \alpha_i X_i\right] = \sum_{i=1}^n \alpha_i \mathbb{E}[X_i].$$

Proof. Exchange the two finite sums:

$$\mathbb{E}[X] = \sum_{\omega} \left(\sum_{i=1}^n \alpha_i X_i(\omega) \right) \mathbb{P}(\omega) = \sum_{i=1}^n \alpha_i \sum_{\omega} X_i(\omega) \mathbb{P}(\omega) = \sum_{i=1}^n \alpha_i \mathbb{E}[X_i]. \quad \blacksquare$$

Remark 2.1.3. This holds for *any* random variables X_i ; no independence is needed. The proof only ever looks at one outcome ω at a time, so it does not care how the X_i are related to each other. This is what makes linearity so powerful: the pieces X_i can be as dependent as they like, as in both examples below.

The pieces we use are almost always indicators.

Definition 2.1.4 (Indicator). The **indicator random variable** of an event E is

$$\mathbb{1}_E = \begin{cases} 1, & \text{if } E \text{ occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

It satisfies $\mathbb{E}[\mathbb{1}_E] = 1 \cdot \mathbb{P}(E) + 0 \cdot \mathbb{P}(E^c) = \mathbb{P}(E)$.

So if X counts how many of the events $E_1, \dots, E_m$ occur, then $X = \sum_i \mathbb{1}_{E_i}$, and by [Theorem 2.1.2](#)

$$\mathbb{E}[X] = \sum_{i=1}^m \mathbb{E}[\mathbb{1}_{E_i}] = \sum_{i=1}^m \mathbb{P}(E_i).$$

The expected number of events that occur is the sum of their probabilities, whatever the dependencies between them.

§2.1.iii. Two First Examples

The first example shows that the union bound arguments of the previous chapter are a special case of the first moment principle.

Example 2.1.5 (Union bound). Let $E_1, E_2, \dots, E_m$ be bad events, e.g. in Erdős's lower bound for $R(k)$, the events that a given k -set of vertices spans a monochromatic clique in a random coloring of K_n . Let

$$X = \#\{\text{bad events that occur}\} = \sum_{i=1}^m \mathbb{1}_{E_i} \Rightarrow \mathbb{E}[X] = \sum_{i=1}^m \mathbb{E}[\mathbb{1}_{E_i}] = \sum_{i=1}^m \mathbb{P}(E_i).$$

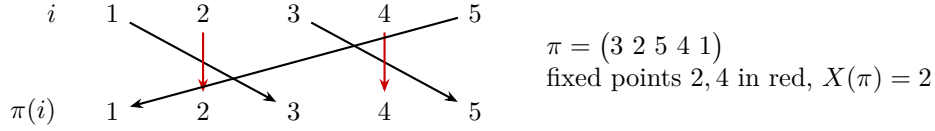
If $\sum_i \mathbb{P}(E_i) < 1$, then $\mathbb{E}[X] < 1$, and by [Lemma 2.1.1](#), with positive probability $X \leq \mathbb{E}[X] < 1$. As X takes values in $\{0, 1, 2, \dots\}$, $X < 1 \Leftrightarrow X = 0$, i.e. with positive probability no bad event occurs. This is exactly the conclusion of the union bound.

In fact the union bound itself follows. Since X is a non-negative integer, $\mathbb{1}_{\{X \geq 1\}} \leq X$ pointwise, and taking expectations,

$$\mathbb{P}\left(\bigcup_{i=1}^m E_i\right) = \mathbb{P}(X \geq 1) = \mathbb{E}[\mathbb{1}_{\{X \geq 1\}}] \leq \mathbb{E}[X] = \sum_{i=1}^m \mathbb{P}(E_i).$$

Example 2.1.6 (Fixed points of permutations). Let $\pi \in S_n$ be a uniformly random permutation of $[n]$, and let

$$X(\pi) = \#\{\text{fixed points of } \pi\} = \#\{i \in [n] : \pi(i) = i\}.$$



The distribution of X is complicated (already $\mathbb{P}(X = 0)$ needs inclusion–exclusion), but we only want its mean. Write X as a sum of indicators, one per position:

$$X = \sum_{i=1}^n \mathbb{1}_{\{i \text{ is a fixed point}\}}.$$

For each i , $\pi(i)$ is uniform on $[n]$, so $\mathbb{P}(\pi(i) = i) = \frac{(n-1)!}{n!} = \frac{1}{n}$. By linearity,

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{P}(i \text{ is a fixed point}) = \sum_{i=1}^n \frac{1}{n} = 1,$$

for every n . Note that the events $\{\pi(i) = i\}$ are not independent: if $n - 1$ of them occur, so does the last one. Linearity does not notice.

§2.2. Sum-Free Sets

The first real application of the first moment principle is to a problem in additive combinatorics.

§2.2.i. The Problem

Definition 2.2.1 (Sum-free set). A **sum** is a triple of integers (x, y, z) s.t. $x + y = z$. A set $A \subseteq \mathbb{Z}$ is **sum-free** if there is no sum in A^3 , i.e. there are no $x, y, z \in A$ with $x + y = z$.

Here x, y, z need not be distinct: a sum-free set cannot contain both x and $2x$, and it never contains $0 = 0 + 0$. The same definition makes sense in any abelian group, and we will use it in $\mathbb{Z}_p$.

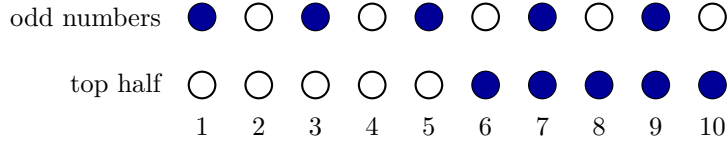
Definition 2.2.2. Given a finite set $S \subseteq \mathbb{Z}$, let

$$f(S) = \max\{|A| : A \subseteq S \text{ is sum-free}\}$$

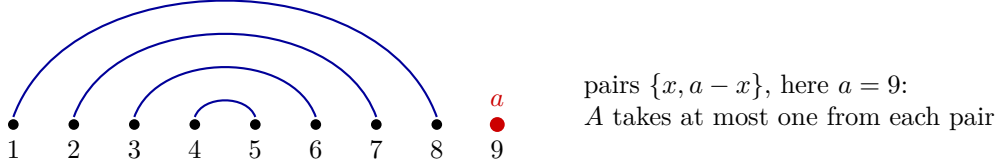
be the size of a largest sum-free subset of S .

Example 2.2.3 (Warm-up: $f([n]) = \lceil \frac{n}{2} \rceil$). *Lower bound.* Two sum-free subsets of $[n]$ of size $\lceil \frac{n}{2} \rceil$ are

- the odd numbers, since odd + odd is even;
- the numbers $> \frac{n}{2}$, since the sum of two of them exceeds n and so is not in the set.



Upper bound. Let $A \subseteq [n]$ be sum-free and $a = \max A$. For every $x < a$, at most one of x and $a - x$ lies in A , since $x + (a - x) = a \in A$. The pairs $\{x, a - x\}$ cover $[a - 1]$:



If a is odd there are $\frac{a-1}{2}$ pairs, so $|A| \leq \frac{a-1}{2} + 1 = \frac{a+1}{2}$. If a is even there are $\frac{a-2}{2}$ pairs and the middle element $\frac{a}{2}$ is excluded on its own, as $\frac{a}{2} + \frac{a}{2} = a$, so $|A| \leq \frac{a-2}{2} + 1 = \frac{a}{2}$. Either way $|A| \leq \frac{a+1}{2} \leq \frac{n+1}{2}$, i.e. $|A| \leq \lceil \frac{n}{2} \rceil$.

So an interval always contains a sum-free half. The interval is very structured, though, and both constructions used that structure. What if the numbers are arbitrary?

Question 2.2.4. Does *every* set of n integers contain a large sum-free subset?

Definition 2.2.5. For $n \in \mathbb{N}$, let¹

$$f(n) = \min\{f(S) : S \subseteq \mathbb{Z} \setminus \{0\}, |S| = n\}.$$

Remark 2.2.6 (Why the minimum). The maximum is trivially n : take S itself sum-free, e.g. $a, b > 2a, 2b + 1, 4b + 3, \dots$ with $a > 0$, each element larger than twice the previous one, hence not a sum of two earlier ones.²

Observation . $f(n) \leq f([n]) = \lceil \frac{n}{2} \rceil$.

Now S may be any set of non-zero integers, including negative ones, with no structure to exploit. Yet a constant fraction of it can always be kept.

Theorem 2.2.7 (Erdős, 1965)

$f(n) \geq \frac{n+1}{3}$. That is, every set of n non-zero integers contains a sum-free subset of size at least $\frac{n+1}{3}$.

Proof. Let $S \subseteq \mathbb{Z} \setminus \{0\}$ with $|S| = n$.

Lemma 2.2.8

There are infinitely many primes $p \equiv 2 \pmod{3}$.

Proof. Given such primes $p_1, \dots, p_m$, let $N = 3p_1 \cdots p_m - 1 \equiv 2 \pmod{3}$. Then $3 \nmid N$, and not all prime factors of N are $\equiv 1 \pmod{3}$, else $N \equiv 1$. So some prime $q \equiv 2 \pmod{3}$ divides N , and $q \neq p_i$ since $p_i \mid N + 1$. ■

¹We exclude 0, which can never lie in a sum-free set.

² $x = y$ is allowed, so beating $a + b$ is not enough: $1 + 1 = 2$.

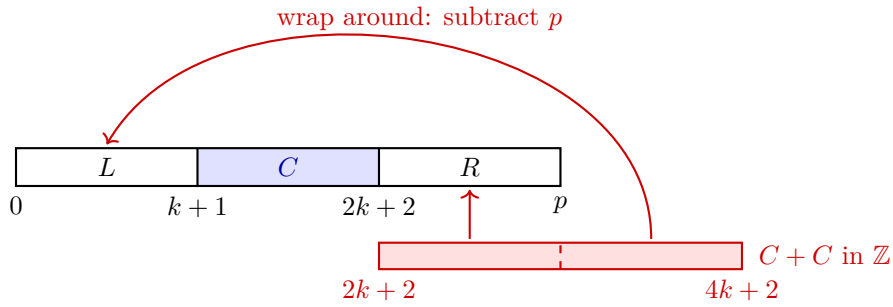
By Lemma 2.2.8, choose a prime $p = 3k + 2$ with $p > \max_{s \in S} |s|$, so $p \nmid s$ for all $s \in S$. We identify $\mathbb{Z}_p$ with $\{0, 1, \dots, p-1\}$, and for an integer m write $m \bmod p \in \{0, 1, \dots, p-1\}$ for its remainder. Let

$$C = \{k+1, k+2, \dots, 2k+1\}.$$

Lemma 2.2.9

C is sum-free in $\mathbb{Z}_p$, and $\frac{|C|}{p-1} = \frac{k+1}{3k+1} > \frac{1}{3}$.

Proof. For $c_1, c_2 \in C$ we have $2k+2 \leq c_1 + c_2 \leq 4k+2$. If $c_1 + c_2 \leq 3k+1$, then $(c_1 + c_2) \bmod p = c_1 + c_2 \in R = \{2k+2, \dots, 3k+1\}$; otherwise $(c_1 + c_2) \bmod p = c_1 + c_2 - p \in L = \{0, \dots, k\}$. Either way $(c_1 + c_2) \bmod p \notin C$. Finally $|C| = k+1$ and $p-1 = 3k+1$. ■



Let x be uniformly random in $\{1, 2, \dots, p-1\}$, i.e. $\mathbb{P}(x = a) = \frac{1}{p-1}$ for each a , and

$$A_x = \{s \in S : xs \bmod p \in C\} \subseteq S.$$

We show that every A_x is sum-free, and that $|A_x| \geq \frac{n+1}{3}$ for some x ; this A_x then proves the theorem.

A_x is sum-free: if $s_1 + s_2 = s_3$ with $s_1, s_2, s_3 \in A_x$, then $xs_1 + xs_2 = xs_3$, so $(xs_1 \bmod p) + (xs_2 \bmod p) \equiv xs_3 \bmod p \pmod{p}$ with all three remainders in C , contradicting Lemma 2.2.9.

Fix $s \in S$. We claim that $xs \bmod p$ takes each value $1, 2, \dots, p-1$ with probability exactly $\frac{1}{p-1}$. Consider

$$\varphi_s: \{1, \dots, p-1\} \rightarrow \{1, \dots, p-1\}, \quad a \mapsto as \bmod p.$$

- It is well defined, i.e. $as \bmod p \neq 0$: p is prime and $p \nmid a$, $p \nmid s$, so $p \nmid as$.
- It is injective: if $as \bmod p = a's \bmod p$, then $p \mid (a - a')s$, so $p \mid a - a'$ as $p \nmid s$; since $|a - a'| < p$, $a = a'$.
- An injection of a finite set into itself is a bijection.

So for each $y \in \{1, \dots, p-1\}$ there is exactly one a with $as \bmod p = y$, and

$$\mathbb{P}(xs \bmod p = y) = \mathbb{P}(x = \varphi_s^{-1}(y)) = \frac{1}{p-1}.$$

Summing over $y \in C$,

$$\mathbb{P}(xs \bmod p \in C) = \sum_{y \in C} \frac{1}{p-1} = \frac{|C|}{p-1} = \frac{k+1}{3k+1} > \frac{1}{3}.$$

Writing $|A_x| = \sum_{s \in S} \mathbb{1}_{\{xs \bmod p \in C\}}$, linearity of expectation gives

$$\mathbb{E}[|A_x|] = \sum_{s \in S} \mathbb{P}(xs \bmod p \in C) > \frac{n}{3}.$$

By [Lemma 2.1.1](#) some x has $|A_x| > \frac{n}{3}$, and as $|A_x|$ is an integer, $|A_x| \geq \frac{n+1}{3}$. ■

Remark 2.2.10. So $\frac{n+1}{3} \leq f(n) \leq \lceil \frac{n}{2} \rceil$. Bourgain (1997) improved the lower bound to $\frac{n+2}{3}$, and Eberhard, Green and Manners (2014) showed that the constant $\frac{1}{3}$ cannot be improved: $f(n) \leq (\frac{1}{3} + o(1))n$. So for arbitrary sets the random dilation is essentially optimal, and the interval $[n]$, with its sum-free half, is far from the worst case.